

Probability

Notes taken by Runqiu Ye
Lectures by Konstantin Tikhomirov
Carnegie Mellon University

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1 Measure theory review

1.1 Measurable space and mapping

Definition (σ -field). A collection of subsets $\Sigma \subset 2^\Omega$ is a σ -field if

- $\emptyset \in \Sigma$.
- If $A \in \Sigma$, then $A^c \in \Sigma$.
- If $\{A_i\}_{i=1}^\infty \subset \Sigma$, then $\bigcup_{i=1}^\infty A_i \in \Sigma$.

The pair (Ω, Σ) is called a measurable space.

Definition (atom). Let Σ be a σ -field. Say $A \in \Sigma$ is an atom if for all $B \in \Sigma$ either $A \subset B$ or $A \cap B = \emptyset$.

Proposition. For all $\omega \in \Omega$, there exists atom $A \in \Sigma$ containing ω if Ω is finite or countable.

Proof. Define $\tilde{A} = \bigcap \{B \in \Sigma : \omega \in B\}$. We can check that $\tilde{A} \in \Sigma$ and $\tilde{A}$ is an atom containing ω . □

Corollary. If Ω is finite or countable, there exists a partition $\Omega = \bigsqcup_i \Omega_i$, where each Ω_i is an atom of Σ . With this partition, Σ is just the power set with respect to $\{\Omega_i\}_i$.

Definition. If $F \subset 2^\Omega$, then the σ -field generated by F is the smallest σ -field containing all elements of F . Write this σ -field as $\sigma(F)$.

Example. Let $\Omega = \{1, 2, 3, 4, 5\}$ and $F = \{\{2, 3\}, \{3, 4\}\}$. Construct σ -field Σ generated by F . Σ is all possible union of sets from the collection $\{\{2\}, \{3\}, \{4\}, \{1, 5\}\}$.

Definition (measurable mapping). Given two measurable spaces (Ω, Σ) and $(\tilde{\Omega}, \tilde{\Sigma})$. Then $f : \Omega \rightarrow \tilde{\Omega}$ is measurable if $f^{-1}(B) \in \Sigma$ for all $B \in \tilde{\Sigma}$.

Definition (Borel σ -field). Let (T, τ) be a topological space. Then the Borel σ -field $\mathcal{B}(T, \tau)$ is defined as the smallest σ -field containing all open sets.

Definition (product measurable space). Given two measurable spaces (Ω, Σ) and $(\tilde{\Omega}, \tilde{\Sigma})$. We can define the product measurable space as follows: let the ground set be $\Omega \times \tilde{\Omega}$, and let $\Sigma \otimes \tilde{\Sigma}$ be the smallest σ -field containing all rectangles $B \times \tilde{B}$ where $B \in \Sigma$ and $\tilde{B} \in \tilde{\Sigma}$.

More generally, let Λ be an index set and $(\Omega_\lambda, \Sigma_\lambda)_{\lambda \in \Lambda}$. Define the product σ -field $\bigotimes_{\lambda \in \Lambda} \Sigma_\lambda$ be the smallest σ -field containing all elements in the form of $\prod_{\lambda \in \Lambda} B_\lambda$ where $B_\lambda \in \Sigma_\lambda$ and $B_\lambda = \Omega_\lambda$ for all but countably many indices.

Proposition. Let $(\Omega_i, \Sigma_i)_{i=1}^n$ be measurable spaces and $(\prod_{i=1}^n \Omega_i, \bigotimes_{i=1}^n \Sigma_i)$ be the product space. Let (Ω, Σ) be the domain and $f = (f_1, \dots, f_n) : (\Omega, \Sigma) \rightarrow (\prod_{i=1}^n \Omega_i, \bigotimes_{i=1}^n \Sigma_i)$. Suppose f is measurable, then every coordinate projection $f_i : \Omega \rightarrow \Omega_i$ is measurable.

This is also true for arbitrary index set.

Proposition. If $f : (\Omega, \Sigma) \rightarrow (\Omega_f, \Sigma_f)$ and $g : (\Omega, \Sigma) \rightarrow (\Omega_g, \Sigma_g)$, then the concatenation (f, g) is measurable w.r.t. the product space $(\Omega_f \times \Omega_g, \Sigma_f \otimes \Sigma_g)$.

Proof. Let $A \times B$ be such that $A \in \Sigma_f$ and $B \in \Sigma_g$. Then the preimage

$$(f, g)^{-1}(A \times B) = f^{-1}(A) \cap g^{-1}(B) \in \Sigma.$$

By definition, the product σ -field is generated by rectangles, so the proof is complete. $\square$

1.2 Measure space

Definition (measure). Let (Ω, Σ) be a measurable space. Then $\mu : \Sigma \rightarrow [0, \infty]$ is a measure if

- $\mu(\emptyset) = 0$.
- If $A_i \in \Sigma$ is pairwise disjoint then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

Proposition (continuity of measure). If $A_1 \subset A_2 \subset \dots$ is a nested sequence of elements of Σ and μ be any measure on (Ω, Σ) . Then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i).$$

If $A_1 \supset A_2 \supset \dots$ is a nested sequence of elements of Σ and $\mu(A_n) < \infty$ for some n . Then

$$\mu\left(\bigcap_{i=1}^{\infty} A_i\right) = \lim_{i \rightarrow \infty} \mu(A_i).$$

Definition. Let (Ω, Σ, μ) be a measure space.

Say μ is σ -finite if there is a representation $\Omega = \bigcup_{i=1}^{\infty} \Omega_i$ where $\Omega_i \in \Sigma$ and $\mu(\Omega_i) < \infty$.

Say μ is a probability measure if $\mu(\Omega) = 1$.

Definition (completion of measure space). Let (Ω, Σ, μ) be a measure space. Let

$$\tilde{\Sigma} = \{A \cup B : A \in \Sigma, B \subset \Omega, \text{there exists } C \in \Sigma \text{ with } \mu(C) = 0 \text{ and } B \subset C\}.$$

We can check $\tilde{\Sigma}$ is a σ -field. If $\tilde{\mu}$ is a measure on $(\Omega, \tilde{\Sigma})$ which agrees with μ on Σ , then $(\Omega, \tilde{\Sigma}, \tilde{\mu})$ is called a completion of (Ω, Σ, μ) .

1.3 π - λ theorem

Definition (π -system). Let Ω be a set and $\mathcal{P}$ be a collection of subsets of Ω . Then $\mathcal{P}$ is a π -system if it is closed with respect to taking finite intersections. That is, $A, B \in \mathcal{P}$ implies $A \cap B \in \mathcal{P}$.

Example. On the real line $\mathbb{R}$, both $\mathcal{P}_1 = \{(a, b) : a < b\}$ and $\mathcal{P}_2 = \{(-\infty, a] : a \in \mathbb{R}\}$ are π -systems.

Definition (λ -system). Let Ω be a set and $\mathcal{L}$ be a collection of subsets of Ω . Say $\mathcal{L}$ is a λ -system if

- $\emptyset \in \mathcal{L}$.
- $A \in \mathcal{L}$ implies $A^c \in \mathcal{L}$.
- for all countable collection of disjoint elements $A_i \in \mathcal{L}$, we have $\bigcup_{i=1}^{\infty} A_i \in \mathcal{L}$.

For an alternative definition, say $\mathcal{L}$ is a λ -system if

- $\Omega \in \mathcal{L}$.
- If $A, B \in \mathcal{L}$ and $A \subset B$, then $B \setminus A \in \mathcal{L}$.
- If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem (π - λ theorem). Let Ω be a set, $\mathcal{P}$ be a π -system and $\mathcal{L}$ be a λ -system. Also suppose $\mathcal{P} \subset \mathcal{L}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Proof. Let $\ell(\mathcal{P})$ be the smallest λ -system on Ω containing $\mathcal{P}$. The goal is to show that $\ell(\mathcal{P})$ is a σ -field. We need to show that if $A_i \in \ell(\mathcal{P})$ for $1 \leq i < \infty$, then $\bigcup_{i=1}^{\infty} A_i \in \ell(\mathcal{P})$. Note that

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \left(A_i \setminus \bigcup_{j=1}^{i-1} A_j \right),$$

so it suffices to show that $A, B \in \ell(\mathcal{P})$ implies $A \cap B \in \ell(\mathcal{P})$.

For $A \in \ell(\mathcal{P})$ we define

$$W_A = \{B \subset \Omega : A \cap B \in \ell(\mathcal{P})\}.$$

It can be directly verified that W_A is a λ -system.

Take $A \in \mathcal{P}$, then for any $B \in \mathcal{P}$ we have $A \cap B \in \mathcal{P} \subset \ell(\mathcal{P})$. Hence, $\mathcal{P} \subset W_A$ and thus $\ell(\mathcal{P}) \subset W_A$ for all $A \in \mathcal{P}$, as $\ell(\mathcal{P})$ is the smallest λ -system on Ω containing $\mathcal{P}$. Now take $A \in \ell(\mathcal{P})$, we have $A \in W_B$ for all $B \in \mathcal{P}$. It follows that $A \cap B \in \ell(\mathcal{P})$ and thus $B \in W_A$. Hence similarly $\ell(\mathcal{P}) \subset W_A$ for all $A \in \ell(\mathcal{P})$.

Now for any pair $B, C \in \ell(\mathcal{P})$, we have $C \in W_B$ and thus $B \cap C \in \ell(\mathcal{P})$. This completes the proof. □

1.4 Extension theorems

Definition (semi-field). A collection of subsets $S \subset 2^\Omega$ is a semi-field if

- $\emptyset \in S$ and $\Omega \in S$.
- $A, B \in S$ implies $A \cap B \in S$.
- If $A \in S$, then A^c is a finite disjoint union of sets in S .

Theorem (Caratheodory's extension theorem). Let S be a semi-field and let μ be a non-negative function on S satisfying:

- $\mu(\emptyset) = 0$.
- If $A_1, \dots, A_n$ are disjoint and $\bigcup_{i=1}^n A_i \in S$, then $\mu(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mu(A_i)$.
- If $A_1, A_2, \dots$ are such that $\bigcup_{i=1}^\infty A_i \in S$, then $\mu(\bigcup_{i=1}^\infty A_i) \leq \sum_{i=1}^\infty \mu(A_i)$.

Then μ admits a unique extension $\bar{\mu}$ which is a measure on $\bar{S}$, the field (algebra) generated by S . Moreover, if $\bar{\mu}$ is σ -finite then $\bar{\mu}$ admits a unique extension $\tilde{\mu}$ to $\sigma(S)$.

Notation. Let T be any set. Write

$$\mathbb{R}^T = \{(\omega_t)_{t \in T} : \omega_t \in \mathbb{R}\}.$$

Also write $\mathcal{R}^T$ as the σ -field generated by rectangles of the form $\prod_{t \in T} I_t$, where for each $t \in T$, I_t is either a semi-open interval of the form $(a, b]$ with $a < b$ or $I_t = \mathbb{R}$, and $I_t = \mathbb{R}$ for all but finitely many $t \in T$.

Theorem (Kolmogorov's extension theorem). For each finite non-empty subset $J \subset T$, let μ_J be a Borel probability measure in $\mathbb{R}^J$, and assume that the measures $(\mu_J)_{J \subset T, |J| < \infty}$ are compatible, in the sense that whenever $J_1 \subset J_2 \subset T$ with $0 \leq |J_1| \leq |J_2| < \infty$, $I_j \subset \mathbb{R}$ with $j \in J_1$ are Borel subsets of $\mathbb{R}$, and

$$\tilde{I}_j = \begin{cases} I_j & (j \in J_1) \\ \mathbb{R} & (j \in J_2 \setminus J_1), \end{cases}$$

one has

$$\mu_{J_2} \left(\prod_{j \in J_2} \tilde{I}_j \right) = \mu_{J_1} \left(\prod_{j \in J_1} I_j \right).$$

Then there exists a unique probability measure μ on $(\mathbb{R}^T, \mathcal{R}^T)$ consistent with $(\mu_J)_{J \subset T, |J| < \infty}$. That is, one has

$$\mu \left(\prod_{t \in T} I_t \right) = \mu_J \left(\prod_{j \in J} I_j \right)$$

whenever $J \subset T$ with $|J| < \infty$ and $I_t = \mathbb{R}$ for all $t \notin J$.

1.5 Lebesgue Integration

Here we provide a proof for dominated convergence theorem that uses the truncation technique, which will be a useful technique later in the course.

Theorem (dominated convergence theorem). Let $\{f_n\}_{n=1}^\infty$ be a sequence of measurable functions on (Ω, Σ, μ) and $g \geq 0$ be another measurable function. Suppose

1. $\int g d\mu < \infty$.
2. $|f_n|(\omega) \leq g(\omega)$ for all $\omega \in \Omega$ and $n \geq 1$.
3. $f_n \rightarrow f$ pointwise.

Then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu.$$

Proof. **Claim 1.** If h is a function on (Ω, Σ, μ) with $h \geq 0$ and $\int h d\mu < \infty$. Let $\{A_n\}_{n=1}^\infty$ be any sequence of elements of Σ with $\mu(A_n) \rightarrow 0$. Then

$$\int_{A_n} h d\mu \rightarrow 0.$$

Proof of claim. WLOG assume $\mu(A_n) \leq 2^{-n}$ for all n . Define $h_n = h \mathbf{1}_{\bigcup_{i=n}^\infty A_i}$. We then have

1. The sequence $\{h_n\}_{n=1}^\infty$ is monotone.
2. h_n converges to 0 almost everywhere.

Monotone convergence theorem then implies $\lim_{n \rightarrow \infty} \int h_n d\mu = 0$. Meanwhile,

$$0 \leq \int_{A_n} h d\mu \leq \int h_n d\mu,$$

and the proof is complete.

Claim 2. Suppose $h \geq 0$ and $\int h d\mu < \infty$. Let $\{\varepsilon_n\}_{n=1}^\infty$ be a sequence of strictly positive numbers converging to zero. Define

$$B_n = \{\omega \in \Omega : h(\omega) \leq \varepsilon_n\} \in \Sigma.$$

Then

$$\int_{B_n} h d\mu \rightarrow 0.$$

Proof of this claim is left as an exercise.

Now we prove the theorem. Fix $\varepsilon > 0$. By the previous two claims, there exists $M > 0$ and $\delta > 0$ such that

$$\int_{\{g \geq M\}} g d\mu < \varepsilon, \quad \int_{\{g \leq \delta\}} g d\mu < \varepsilon.$$

Let $U = \{\omega : \delta < g(\omega) < M\}$. Since g is integrable, $\mu(U) < \infty$. For $\omega \in U$, let $n_\varepsilon(\omega)$ be the smallest index such that $n \geq n_\varepsilon(\omega)$ implies $|f_n(\omega) - f(\omega)| \leq \varepsilon \mu(U)^{-1}$. It follows that there exists N such that

$$\mu(\{\omega \in U : n_\varepsilon(\omega) > N\}) \leq \frac{\varepsilon}{M}.$$

Then, for $n \geq N$, we have

$$\left| \int_U (f_n - f) d\mu \right| \leq \int_{n_\varepsilon(\omega) \leq N} |f_n - f| d\mu + \int_{n_\varepsilon(\omega) > N} |f_n - f| d\mu \leq 3\varepsilon.$$

Now for $n \geq N$, we have

$$\left| \int (f_n - f) d\mu \right| \leq 3\varepsilon + \int_{U^c} |f - f_n| d\mu \leq 3\varepsilon + 2 \int_{U^c} g d\mu \leq 7\varepsilon.$$

□

Theorem (Markov-Chebyshev inequality). Suppose we have probability measure space (Ω, Σ, P) and $f \geq 0$. Suppose also $\int f dP < \infty$. Then

$$P(\{\omega : f(\omega) > t\}) \leq \frac{1}{t} \int f dP.$$

for all $t > 0$.

Remark. Let $1 \leq p < \infty$. Suppose $f : (\Omega, \Sigma, P) \rightarrow [0, \infty]$ and $\int f^p dP < \infty$. Then

$$P(\{\omega : f(\omega) > t\}) \leq \frac{1}{t^p} \int f^p dP.$$

for all $t > 0$.

Remark. Suppose $\int e^{\lambda f} dP < \infty$ for all $\lambda \in \mathbb{R}$ and $f : (\Omega, \Sigma, P) \rightarrow [0, \infty]$. Then

$$P(\{\omega : f(\omega) > t\}) \leq \frac{1}{e^{\lambda t}} \int e^{\lambda f} dP$$

for all $t > 0$ and $\lambda > 0$.

Theorem (Hölder inequality). Let $p, q \in [1, \infty]$ and $p^{-1} + q^{-1} = 1$. Let (Ω, Σ, μ) be a probability space. For any measurable functions f, g , we have

$$\int |fg| dP \leq \left(\int |f|^p dP \right)^{1/p} \left(\int |g|^q dP \right)^{1/q}.$$

Theorem (Jensen's inequality). Let (Ω, Σ, P) be a probability space and f be integrable. Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex and suppose $\varphi(\infty) = \lim_{x \rightarrow \infty} \varphi(x)$ and $\varphi(-\infty) = \lim_{x \rightarrow -\infty} \varphi(x)$. Then

$$\varphi \left(\int f dP \right) \leq \int \varphi(f) dP.$$

1.6 Product measures and Fubini theorem

Let $(\Omega_1, \Sigma_1, \mu_1)$, $(\Omega_2, \Sigma_2, \mu_2)$ be σ -finite measure spaces. We already defined the product $\Sigma_1 \otimes \Sigma_2$. To define a product measure, we first consider the algebra of rectangles

$$S = \{A \in \Sigma_1 \otimes \Sigma_2 : A = A_1 \times A_2 \text{ for some } A_1 \in \Sigma_1, A_2 \in \Sigma_2\}.$$

Then we can define $\mu = \mu_1 \times \mu_2$ on S by

$$\mu(A) = \mu_1(A_1)\mu_2(A_2)$$

for $A = A_1 \times A_2$. We can check that the definition is self-consistent. That is, if $A = A_1 \times A_2$ is a countable union of disjoint rectangles $\{A_1^{(j)} \times A_2^{(j)}\}_{j=1}^\infty$, we have

$$\mu(A_1 \times A_2) = \sum_{j=1}^\infty \mu(A_1^{(j)} \times A_2^{(j)}).$$

This can be verified with monotone convergence theorem. Now μ is a premeasure and can be uniquely extended to $\Sigma_1 \otimes \Sigma_2$.

Theorem (Fubini-Tonelli). Let $(\Omega_1, \Sigma_1, \mu_1)$, $(\Omega_2, \Sigma_2, \mu_2)$ be σ -finite measure spaces and let (Ω, Σ, μ) be the product space. Suppose f is measurable on the product space. Suppose either f is non-negative or $\int_\Omega |f| d\mu < \infty$. Then

- $y \mapsto f(x, y)$ is Σ_2 measurable for all $x \in \Omega_1$.
- $x \mapsto \int_{\Omega_2} f(x, y) d\mu_2(y)$ is Σ_1 measurable.
- We have

$$\int_{\Omega_1} \int_{\Omega_2} f(x, y) d\mu_2(y) d\mu_1(x) = \int_\Omega f(x, y) d\mu(x, y).$$

Proof. First suppose $f = \mathbf{1}_A$ for $A \in \Sigma$. Also suppose μ_1, μ_2 are finite. Define section

$$A_x = \{y \in \Omega_2 : (x, y) \in A\}.$$

The goal is to show that $A_x \in \Sigma_2$ for all $x \in \Omega_1$. Define a family of sets

$$\mathcal{F}_x = \{B \in \Sigma : B_x \text{ is } \Sigma_2\text{-measurable}\}.$$

It can be verified that $\mathcal{F}_x$ is a σ -field for all $x \in \Omega_1$. Also, $\mathcal{F}_x$ contains all rectangles and thus $\Sigma \subset \mathcal{F}_x$. Hence, we have shown that $y \mapsto \mathbf{1}_A(x, y) = \mathbf{1}_{A_x}(y)$ is measurable for all $x \in \Omega_1$.

Next we show $x \mapsto \mu_2(A_x)$ is measurable and its integral over Ω_1 is equal to $\mu(A)$. Define

$$\mathcal{U} = \left\{ B \in \Sigma : x \mapsto \mu_2(B_x) \text{ is } \Sigma_1\text{-measurable and } \int_{\Omega_1} \mu_2(B_x) d\mu_1 = \mu(B) \right\}$$

It can be verified that $\mathcal{U}$ is a λ -system. Note that $\mathcal{U}$ also contains all rectangles in Σ . It follows that $\mathcal{U} = \Sigma$ and the proof for indicator functions are complete.

Then use linearity to extend to simple functions, and use monotone convergence theorem to prove the statement for non-negative functions. For the case where f is integrable, consider the positive and negative part about f to complete the proof.

□

2 Probability theory basics

2.1 Distributions and densities

Definition. Let $F : \mathbb{R} \rightarrow [0, 1]$. Suppose F is

- right-continuous.
- non-decreasing.
- $\lim_{t \rightarrow -\infty} F(t) = 0$ and $\lim_{t \rightarrow \infty} F(t) = 1$.

Then F is a cumulative distribution function (CDF).

Remark. If we want to define CDF in $\mathbb{R}^2$ then the axioms are

- right-continuous: $F(\tilde{s}, \tilde{t}) \rightarrow F(s, t)$ as $t \downarrow \tilde{t}$ and $s \downarrow \tilde{s}$.
- coordinate-wise non-decreasing.
- $\lim_{s, t \rightarrow \infty} F(s, t) = 1$, $\lim_{s \rightarrow -\infty} F(s, t) = 0$ for any t , and $\lim_{t \rightarrow -\infty} F(s, t) = 0$ for any s .
- For a rectangle with bottom left vertex (a_1, a_2) and top right vertex (b_1, b_2) ,

$$F(b_1, b_2) - F(b_1, a_2) - F(a_1, b_2) + F(a_1, a_2) \geq 0.$$

Now we can connect the notion of CDF with randomness.

Suppose X -random real-valued variable on (Ω, Σ, P) that is almost everywhere finite. Define

$$F_X(t) = P(X(\omega) \leq t)$$

for $-\infty < t < \infty$. It can be verified that F_X is a CDF.

Conversely, for any CDF F , there exists a probability space (Ω, Σ, P) and a real valued random variable on (Ω, Σ, P) with CDF F .

Definition. If X is random variable on (Ω, Σ, P) real valued and a.e. finite. Then we can define the induced Borel probability measure μ_X on $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ by

$$\mu_X(B) := P(X \in B)$$

for all $B \in \mathcal{B}_{\mathbb{R}}$.

Now suppose μ is any Borel probability measure on $\mathbb{R}$. Consider probability space $(\mathbb{R}, \mathcal{B}_{\mathbb{R}}, \mu)$ and formal identity mapping id on $\mathbb{R}$. Then $\mu_{\text{id}} \equiv \mu$.

Theorem. There is a one-to-one correspondence between the family of CDFs and the family of Borel probability measure on $\mathbb{R}$.

Proof. For any Borel probability measure μ , $F_\mu(t) = \mu((-\infty, t])$ is a valid CDF.

Conversely, for any CDF F , there exists unique probability measure μ_F on $\mathbb{R}$ such that $\mu_F((-\infty, t]) = F(t)$ for all $-\infty < t < \infty$. This is a corollary of Caratheorod extension theorem. For detailed proof see notes or textbook. □

Remark. Suppose $X = (X_1, X_2)$ is a random vector in $\mathbb{R}^2$. We can define

$$F_X(s, t) = P(X_1 \leq s, X_2 \leq t).$$

Corresponding results are also true.

Definition (Probability mass function). Let (Ω, Σ, P) be a probability space and $X : \Omega \rightarrow \mathbb{R}$ be random variable. Suppose there exists $S \subset \mathbb{R}$ countable such that $P(X \in S) = 1$. We can define the probability mass function (PMF) f_X via

$$f_X(t) = P(X = t)$$

for $t \in \mathbb{R}$. Due to the restriction, this gives complete description of the distribution, and we can construct CDF F_X via

$$F_X(t) = \sum_{s \leq t} f_X(s).$$

This sum makes sense since the $f_X(s) = 0$ for all but countably many s . Conversely, we can also reconstruct f_X from a CDF F_X .

Definition (Probability density function). Suppose F is a CDF which is absolutely continuous. That is, there exists Borel measurable non-negative function ρ on $\mathbb{R}$ such that

$$F(t) = \int_{-\infty}^t \rho(s) ds$$

for all $-\infty < t < \infty$. This implies F is almost everywhere differentiable and the derivative is ρ . In this case, say ρ is the density function.

If random variable X is such that F_X is absolutely continuous, then the corresponding ρ_X is the probability density function for X .

Remark. Recall that a Borel σ -finite measure μ on the real line is absolutely continuous w.r.t the Lebesgue measure m on $\mathbb{R}$ if $\mu(A) = 0$ whenever $A \in \mathcal{B}_{\mathbb{R}}$ is Lebesgue null. In this case, Randon-Nikodym theorem implies existence of non-negative Borel measurable function f such that $\mu(A) = \int_A f dm$.

Theorem. Suppose X RV on (Ω, Σ, P) is real-valued and a.e. finite. The following are equivalent:

1. F_X is absolutely continuous.
2. μ_X is absolutely continuous w.r.t. Lebesgue measure.

Moreover, ρ_X is also the derivative of μ_X w.r.t. Lebesgue measure. That is, for any $A \in \mathcal{B}_{\mathbb{R}}$,

$$\mu_X(A) = \int_A \rho_X(t) dt.$$

2.2 Independence

Definition. Say two events $A, B \in \Sigma$ are independent if $P(A \cap B) = P(A)P(B)$.

It is easy to verify that A, B are independent implies A^c, B are independent.

Remark. Suppose $P(B) > 0$, then the conditional probability of A given B is defined as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Then, independence of A and B is equivalent to $P(A|B) = P(A)$.

Definition. Let $A_1, \dots, A_n$ be events. Say they are mutually independent if for any $\emptyset \neq I \subset [n]$, we have

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i).$$

This is equivalent to saying that for every $2 \leq i \leq n$, the event A_i is independent from any event generated by $A_1, \dots, A_{i-1}$, or A_i is independent from $\sigma(A_1, \dots, A_{i-1})$.

Remark. The events $A_1, \dots, A_n$ are called k -wise independent if any k -subset of the events are mutually independent. For $k < n$, this notion is strictly weaker than mutual independence of all n events. As an example, consider P to be the uniform distribution on $\{1, \dots, 4\}$. Let $A_1 = \{1, 2\}$, $A_2 = \{1, 3\}$, and $A_3 = \{2, 3\}$. Then they are pairwise independent but not mutually independent.

Definition. A collection of events $\{A_\lambda\}_{\lambda \in \Lambda}$ on (Ω, Σ, P) are mutually independent if any finite subset of events are mutually independent.

Definition. Let (Ω, Σ, P) be a probability space. Two σ -subfields are independent if for any $A \in \Sigma_1$ and $B \in \Sigma_2$, A, B are independent.

Definition. Let (Ω, Σ, P) be a probability space and X, Y be two real-valued random variables. Say X and Y are independent if

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

for any $A, B \in \mathcal{B}_{\mathbb{R}}$.

Equivalently, let Σ_X, Σ_Y be the σ -field generated by X and Y . Then independence of X and Y is equivalent to independence of Σ_X and Σ_Y .

Now we explore how this connect with product structure.

Proposition. Let $(\Omega_1, \Sigma_1, \mathbb{P}_1)$ and $(\Omega_2, \Sigma_2, \mathbb{P}_2)$ be two probability spaces and let (Ω, Σ, P) be the product space. Let X and Y be two random variables on (Ω, Σ, P) . Suppose there exists some measurable functions such that $X(\omega_1, \omega_2) = g(\omega_1)$, and $Y(\omega_1, \omega_2) = h(\omega_2)$. Then X and Y are independent.

Proof. Let $A, B \in \mathcal{B}_{\mathbb{R}}$. Then

$$\begin{aligned} P(X \in A, Y \in B) &= P((\omega_1, \omega_2) : \omega_1 \in g^{-1}(A), \omega_2 \in h^{-1}(B)) \\ &= P(\{\omega_1 \in g^{-1}(A)\} \times \{\omega_2 \in h^{-1}(B)\}) \\ &= \mathbb{P}_1(\omega_1 \in g^{-1}(A)) \mathbb{P}_2(\omega_2 \in h^{-1}(B)). \end{aligned}$$

However,

$$\begin{aligned} \mathbb{P}_1(\omega_1 \in g^{-1}(A)) &= P((\omega_1, \omega_2) : \omega_1 \in g^{-1}(A), \omega_2 \in \Omega_2) \\ &= P(X \in A), \end{aligned}$$

and similarly for Y .

□

Remark. Let (Ω, Σ, P) be a probability space and suppose X, Y be two random variables that are independent and a.e. finite. They then generate two Borel probability measure μ_X and μ_Y on $\mathbb{R}$. Define a product probability space as of $(\mathbb{R}^2, \mathcal{B}_{\mathbb{R}^2}, \mu_X \times \mu_Y)$. Define $\tilde{X}(x, y) = x$ and $\tilde{Y}(x, y) = y$ as random variables on the product space. By definition, $\tilde{X}$ is equidistributed with X . That is, $\mu_{\tilde{X}} = \mu_X$ and $F_{\tilde{X}} = F_X$. Similarly $\mu_{\tilde{Y}} = \mu_Y$. Also, $\tilde{X}, \tilde{Y}$ are independent. Now (X, Y) and $(\tilde{X}, \tilde{Y})$ have the same distribution.

Remark. If X and Y are independent, then their joint distribution $F_{(X,Y)}$ is uniquely determined by the individual distributions of F_X, F_Y . Indeed,

$$F_{(X,Y)}(s, t) = F_X(s)F_Y(t).$$

Remark. Let (Ω, Σ, P) be a probability space and suppose X, Y be two random variables that are independent. Suppose they have densities ρ_X, ρ_Y , then the distribution density of vector (X, Y) is $\rho_{(X,Y)}(s, t) = \rho_X(s)\rho_Y(t)$.

Remark. If X and Y are independent random variable, and f, g are measurable functions. Then $f(X)$ and $g(Y)$ are independent as well.

Remark. Given probability space (Ω, Σ, P) and random variable X . It may not exists another random variable Y that is independent from X on the same probability space. See the following example.

Example. As an example, consider $([0, 1], \mathcal{B}_{[0,1]}, m)$ and $X(\omega) = \omega$, so X is uniform on $[0, 1]$. The goal is to construct variable Y such that $Y \sim \text{Bernoulli}(\frac{1}{2})$.

Proof. Let $A \in \mathcal{B}_{[0,1]}$ and $t \in [0, 1]$. Define the density of set A at point t to be

$$\lim_{\varepsilon \rightarrow 0} \frac{m(A \cap [t - \varepsilon, t + \varepsilon])}{2\varepsilon}.$$

The Lebesgue density theorem says this is well defined and takes values in $\{0, 1\}$ for m -a.e. $t \in [0, 1]$.

Suppose such Y exists, we can view Y as an indicator of a set $A \subset [0, 1]$ of probability $\frac{1}{2}$. That is, $Y = \mathbf{1}_A$ and $P(A) = \frac{1}{2}$. Choose any point $t \in (0, 1)$ such that density of A at t is well-defined. WLOG assume the density is 1. Pick $\varepsilon > 0$ such that $m(A \cap [t - \varepsilon, t + \varepsilon]) \geq \frac{3}{2}\varepsilon$. Now,

$$\begin{aligned} m(A \cap [t - \varepsilon, t + \varepsilon]) &= P(Y = 1, X \in [t - \varepsilon, t + \varepsilon]) \\ &= P(Y = 1) P(X \in [t - \varepsilon, t + \varepsilon]) \\ &= \varepsilon, \end{aligned}$$

a contradiction.

Alternatively, we can derive a contradiction using $m(A) = P(Y = 1, X \in A)$.

□

Remark. The goal is to have statements “independent” from the underlying probability space.

2.3 Convolution

Definition (convolution). Let μ, ν be Borel probability measures on $\mathbb{R}$. The convolution of μ and ν is a probability measure on $\mathbb{R}$ such that

$$(\mu * \nu)(S) = \int_{\mathbb{R}^2} \mathbf{1}_S(x + y) d(\mu \times \nu)$$

for all $S \in \mathcal{B}_{\mathbb{R}}$.

Remark. Suppose both μ and ν have densities ρ_μ and ρ_ν . That is, $\mu \ll m$ and $\nu \ll m$. It follows that

$$\begin{aligned} (\mu * \nu)(S) &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} \mathbf{1}_S(x + y) \rho_\mu(x) dx \right) \rho_\nu(y) dy \\ &= \int_{\mathbb{R}} \left(\int_S \rho_\mu(w - y) dw \right) \rho_\nu(y) dy \\ &= \int_S \left(\int_{\mathbb{R}} \rho_\mu(w - y) \rho_\nu(y) dy \right) dw \end{aligned}$$

Note that this implies $\mu * \nu \ll m$ and

$$\rho_{\mu * \nu}(w) = \int_{\mathbb{R}} \rho_\mu(w - y) \rho_\nu(y) dy,$$

which is the convolution of function ρ_μ and ρ_ν .

Remark. If X and Y are two independent random variables and μ_X and μ_Y are the corresponding induced Borel measure $\mathbb{R}$, then $\mu_{X+Y} = \mu_X * \mu_Y$.

Remark. Suppose X and Y are independent and their CDF is F_X and F_Y , then

$$F_{X+Y}(t) = \int_{\mathbb{R}} F_X(t-w) d\mu_Y(w) = \int_{\mathbb{R}} F_X(t-w) dF_Y(w).$$

2.4 Moments

In this section we explore the computation and basic properties of moments.

Definition (moments). Let X be a random variable on (Ω, Σ, P) . The p -th absolute moment of X is

$$E|X|^p = \int_{\Omega} |X|^p dP.$$

This is always well-defined but can be infinite.

If p is positive integer, the p -th moment of X is

$$EX^p = \int_{\Omega} X^p dP$$

whenever it is defined.

In particular, EX is the mean or expectation, the variance is $\text{var}(X) = E(X - EX)^2$ whenever the expectation is defined, and the standard deviation is $\sqrt{\text{var}(X)}$.

Proposition. Let X be random variable (Ω, Σ, P) and $0 < p < q < \infty$. Then,

$$(E|X|^p)^{1/p} \leq (E|X|^q)^{1/q}.$$

Proof. Define $Y = |X|^p$, then we want to show that $EY \leq (EY^{q/p})^{p/q}$. Note that $t \mapsto |t|^{q/p}$ is convex. Therefore, by Jensen's inequality,

$$(EY)^{q/p} \leq E(Y^{q/p}).$$

□

Now we want to show that moments only depends on the distribution of the random variable, and it does not carry unnecessary information about the underlying probability space. To do this, we first show the following proposition.

Proposition. Let g be measurable and X a random variable. Then,

$$E|g(X)| = \int_{\mathbb{R}} |g(t)| d\mu_X(t).$$

Moreover, if $E|g(X)| < \infty$, then

$$Eg(X) = \int_{\mathbb{R}} g(t) d\mu_X(t).$$

Corollary. If EX is well-defined, then

$$EX = \int_{\mathbb{R}} t d\mu_X(t).$$

Moreover, if X has distribution density ρ_X , then $EX = \int_{\mathbb{R}} t \rho_X(t) dt$.

Proposition. If $X \geq 0$, then

$$EX = \int_0^\infty P(X \geq t) dt.$$

In particular, if X is non-negative and integer valued, then $EX = \sum_{i=1}^\infty P(X \geq i)$.

Proof. With Fubini-Tonelli, we have

$$EX = \int_0^\infty s d\mu_X(s) = \int_0^\infty \int_0^s 1 dt d\mu_X(s) = \int_0^\infty \int_t^\infty 1 d\mu_X dt = \int_0^\infty P(X \geq t) dt.$$

□

Proposition. Let $X_1, \dots, X_n$ are random variables on (Ω, Σ, P) . Suppose they are either all non-negative or all integrable. Suppose also that they are mutually independent. Then,

$$E\left(\prod_{i=1}^n X_i\right) = \prod_{i=1}^n EX_i.$$

Proof. It suffices to show the statement for $n = 2$. Define independent variables $\widetilde{X}_1$ and $\widetilde{X}_2$ on the product space $(\mathbb{R}^2, \mathcal{B}_{\mathbb{R}^2}, \mu_{X_1} \times \mu_{X_2})$ by coordinate projection. We know $\widetilde{X}_i$ is equidistributed with X_i , so

$$E[X_1 X_2] = E[\widetilde{X}_1 \widetilde{X}_2] = E[\widetilde{X}_1] E[\widetilde{X}_2] = E[X_1] E[X_2],$$

where in the second equality we used Fubini-Tonelli.

□

Remark. Recall that whenever EX is finite, $\text{var } X = E(X - EX)^2 = EX^2 - (EX)^2$. If $X_1, \dots, X_n$ are pairwise independent and have well-defined variances, then

$$\text{var}(X_1 + \dots + X_n) = \sum_{i=1}^n \text{var } X_i.$$

Definition (moment generating function). Let X be a random variable, and $\lambda \in \mathbb{R}$. Define the moment generating function of X via

$$M_X(\lambda) = E \exp(\lambda X).$$

Suppose $M_X(\lambda)$ is finite in some neighborhood of 0. Then,

$$M_X(\lambda) = E \left[\sum_{n=1}^{\infty} \frac{(\lambda x)^n}{n!} \right] = \sum_{n=1}^{\infty} \frac{\lambda^n E X^n}{n!}.$$

It follows that $M'_X(0) = EX$. Now to justify the exchange of integral and summation, note that

$$S_N = \sum_{i=1}^N \frac{(\lambda x)^n}{n!} \rightarrow e^{\lambda x},$$

and

$$|S_N| \leq \sum_{i=1}^N \frac{|\lambda x|^n}{n!} \leq e^{|\lambda x|}.$$

However, $e^{|\lambda x|} \leq e^{\lambda x} + e^{-\lambda x}$. The expectation of RHS is finite for small λ by assumption, so the claim follows from dominated convergence theorem.

Proposition. Let X be a non-negative random variable. Then TFAE:

1. M_X is finite in some neighborhood of 0.
2. $\limsup_{p \rightarrow \infty} p^{-1}(EX^p)^{1/p} < \infty$.

Proof. (1) $\implies$ (2). Suppose $M_X(\varepsilon) = E \left[\sum_{n=1}^{\infty} \frac{\varepsilon^n X^n}{n!} \right] < \infty$. This implies that

$$\sup_{n \geq 1} E \left[\frac{\varepsilon^n X^n}{n!} \right] < \infty.$$

Let $1 \leq C < \infty$ be such that $E \left[\frac{\varepsilon^n X^n}{n!} \right] < C$ for all $n \geq 1$. We then have

$$\left(\frac{\varepsilon}{n!} \right)^{1/n} (EX^n)^{1/n} \leq C^{1/n}$$

It follows from Sterling's formula $n! \sim (n/e)^n$ that

$$\frac{1}{n} (EX^n)^{1/n} \leq C$$

for some other constant C .

(2) $\implies$ (1). Suppose $\limsup_{p \rightarrow \infty} p^{-1} (EX^p)^{1/p} \leq C < \infty$. It follows that for any $n \geq 1$

$$\frac{1}{n!} (EX^n) \leq C^n \frac{n^n}{n!} \leq C^n$$

for some other constant C . Take $\varepsilon = \frac{1}{2C}$, we then have

$$E \left[\frac{\varepsilon^n X^n}{n!} \right] \leq 2^{-n}.$$

Therefore, $M_X(\varepsilon) = E \left[\sum_{n=1}^{\infty} \frac{\varepsilon^n X^n}{n!} \right] < \infty$.

□

Now we present an example of moment method for approximating random variables.

Example. Let $G \sim G(n, \frac{1}{2})$ be the Erdos-Renyi random graph. The goal is to estimate the number of triangles in G . Let N be the number of triangles in G . We have

$$N = \sum_{\substack{S \subset [n] \\ |S|=3}} b_S,$$

where b_S is the indicator that S is a triangle in G . It follows that $EN = \frac{1}{8} \binom{n}{3}$. Also,

$$EN^2 = \sum_{|S|=3} \sum_{|S'|=3} E[b_S b_{S'}]$$

To compute this, we consider several cases.

1. If $S \cap S' = \emptyset$, then $E[b_S b_{S'}] = \frac{1}{64}$.
2. If $|S \cap S'| = 1$, then $E[b_S b_{S'}] = \frac{1}{64}$.
3. If $|S \cap S'| = 3$, then $E[b_S b_{S'}] = \frac{1}{8}$.
4. If $|S \cap S'| = 2$, then $E[b_S b_{S'}] = \frac{1}{32}$.

Hence,

$$EN^2 = \frac{1}{64} \binom{n}{3}^2 \pm O(n^4),$$

where the $O(n^4)$ term comes from the cases where $|S \cap S'| = 2$ or 3 . Therefore,

$$\text{var } N = EN^2 - (EN)^2 = O(n^4).$$

Using Chebyshev's inequality, for each $t > 0$ we have

$$P(|N - EN| \geq t \cdot EN) \leq \frac{\text{var } N}{t^2 (EN)^2} = t^{-2} O(n^{-2}) \rightarrow 0.$$

as $n \rightarrow \infty$.

We have the following proposition on subgaussian decay of moments.

Proposition. Let X be a random variable, TFAE:

1. $\sup_{p \geq 1} (E|X|^p)^{1/p} p^{-1/2} < \infty$.
2. there exists $c > 0$ such that $P(|X| > t) \leq 2 \exp(-ct^2)$ for all $t > 0$.

If any of these statement is satisfied, X is called a *subgaussian variable*.

Proof. Exercise. □

Theorem (Khintchine's inequality). There exists a universal constant $C < \infty$ such that for any $n \in \mathbb{N}$, $a_1, \dots, a_n \in \mathbb{R}$, and $p \geq 1$, we have

$$\left(E \left| \sum_{i=1}^n a_i r_i \right|^p \right)^{1/p} \leq C \sqrt{p} \|a\|_2,$$

where $r_1, \dots, r_n$ are i.i.d. Rademacher variables: $P(r_i = 1) = P(r_i = -1) = \frac{1}{2}$.

Proof. WLOG assume $\|a\|_2 = 1$. Let $\lambda > 0$. For any $t > 0$, we have

$$P\left(\sum_{i=1}^n a_i r_i > t\right) = P\left(\exp\left(\lambda \sum_{i=1}^n a_i r_i\right) > \exp(\lambda t)\right) \leq \frac{E \exp\left(\lambda \sum_{i=1}^n a_i r_i\right)}{\exp(\lambda t)}$$

from Markov's inequality. By independence, we have

$$P\left(\sum_{i=1}^n a_i r_i > t\right) \leq \frac{\prod_{i=1}^n E \exp(\lambda a_i r_i)}{\exp(\lambda t)} = \frac{\prod_{i=1}^n \frac{1}{2} (\exp(\lambda a_i) + \exp(-\lambda a_i))}{\exp(\lambda t)}.$$

Now we use $\exp(x) \leq x + \exp(x^2)$ to obtain

$$P\left(\sum_{i=1}^n a_i r_i > t\right) \leq \frac{\prod_{i=1}^n \exp(\lambda^2 a_i^2)}{\exp(\lambda t)} = \exp\left(\lambda^2 \|a\|_2^2 - \lambda t\right).$$

This holds for any $\lambda > 0$. Substituting $\lambda = \frac{1}{2} t \|a\|_2^{-2}$, we have

$$P\left(\sum_{i=1}^n a_i r_i > t\right) \leq \exp\left(-\frac{1}{4} \frac{t^2}{\|a\|_2^2}\right)$$

Similarly we can bound $P(\sum_{i=1}^n a_i r_i < -t)$ as the Rademacher variables are symmetric. Hence we have

$$P\left(\left|\sum_{i=1}^n a_i r_i\right| > t\right) \leq 2 \exp\left(-\frac{1}{4} \frac{t^2}{\|a\|_2^2}\right).$$

The theorem follows from the previous proposition. □

Theorem (Anti-concentration inequalities). Let X be a non-negative random variable and $1 \leq p < q < \infty$. Suppose $(EX^p)^{1/p} \leq C (EX^q)^{1/q}$ for some constant C . Then there exists $\varepsilon > 0$ depending only on p, q, C such that

$$P\left(X \geq \varepsilon (EX^p)^{1/p}\right) \geq \varepsilon.$$

2.5 Convergence of random variable

Definition. Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables and X is also a random variable. Say X_n converges to X *in distribution* or *weakly* if for any compactly supported continuous function f , we have

$$\int_{\mathbb{R}} f(t) d\mu_{X_n}(t) \rightarrow \int_{\mathbb{R}} f(t) d\mu_X(t).$$

Write $X_n \Rightarrow X$ or $X_n \xrightarrow{w} X$.

Proposition. $X_n \xrightarrow{w} X$ iff $F_{X_n}(t) \rightarrow F_X(t)$ at every point t where F_X is continuous.

Definition. Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables and X is also a random variable defined on the same probability space. Say X_n converges to X *in probability* if for any $\varepsilon > 0$,

$$P(|X_n - X| < \varepsilon) \rightarrow 1.$$

Write $X_n \xrightarrow{p} X$.

Proposition. If $X_n \xrightarrow{p} X$, then $X_n \Rightarrow X$.

Proof. Let $t \in \mathbb{R}$ be such that F_X is continuous at point t and let $\varepsilon > 0$. Let $\delta > 0$ be such that $F_X(t + \delta) \leq F_X(t) + \varepsilon$ and $F_X(t - \delta) \geq F_X(t) - \varepsilon$. Since $P(|X - X_n| < \delta) \rightarrow 1$, there exists n_0 such that $n \geq n_0$ implies

$$P(|X_n - X| < \delta) \geq 1 - \varepsilon.$$

Take $n \geq n_0$. We have

$$\begin{aligned} F_{X_n}(t) &= P(X_n \leq t) \\ &\leq P(|X_n - X| > \delta) + P(X \leq t + \delta) \\ &\leq \varepsilon + F_X(t) + \varepsilon \\ &\leq F_X(t) + 2\varepsilon. \end{aligned}$$

Similarly for the other side. □

Proposition. If $X_n \rightarrow X$ a.s., then $X_n \xrightarrow{p} X$.

Proof. Let $\varepsilon > 0$. For each $\omega \in \Omega$, define

$$n_\varepsilon(\omega) = \min \{n : |X_m(\omega) - X(\omega)| \leq \varepsilon \text{ for all } m \geq n\} < \infty.$$

By continuity of measure, for any $\delta > 0$ there exists N such that $P(n_\varepsilon \leq N) \geq 1 - \delta$. This implies that for all $m \geq N$, we have

$$P(|X_m - X| \leq \varepsilon) \geq 1 - \delta.$$

Since $\delta > 0$ is arbitrary, this completes the proof. □

Proposition. Let $c \in \mathbb{R}$ and suppose $\{X_n\}_{n=1}^\infty$ are random variables on a common probability space (Ω, Σ, P) . Then $X_n \Rightarrow c$ iff $X_n \xrightarrow{p} c$.

Proof. Just need to check that $X_n \Rightarrow c$ implies $X_n \xrightarrow{p} c$. We have

$$F_c(t) = \begin{cases} 1 & \text{if } t \geq c, \\ 0 & \text{otherwise.} \end{cases}$$

Take $\varepsilon > 0$, then $c \pm \varepsilon$ are points of continuity. It follows that $F_{X_n}(c + \varepsilon) \rightarrow 1$ and $F_{X_n}(c - \varepsilon) \rightarrow 0$. This implies that

$$P(|X_n - c| > \varepsilon) \rightarrow 0.$$

□

Next we consider algebraic operations and convergence. It is clear that if $X_n \rightarrow X$ and $Y_n \rightarrow Y$ a.e. then $X_n + Y_n \rightarrow X + Y$ a.e. Also, if $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, then $X_n + Y_n \xrightarrow{p} X + Y$. But for convergence in distribution, the same statement need not be true without any extra assumptions. However, suppose X_n and Y_n are independent for each n , X and Y are independent, $X_n \Rightarrow X$, and $Y_n \Rightarrow Y$. Then $X_n + Y_n \Rightarrow X + Y$.

Example. Suppose $\{a_i\}_{i=1}^\infty$ are i.i.d. random variables and

$$\varepsilon_n \sum_{i=1}^n a_i \Rightarrow X$$

for some random variable X and sequence of positive numbers $\{\varepsilon_n\}_{n=1}^\infty$. Then we have

$$\varepsilon_{2n} \sum_{i=1}^{2n} a_i \Rightarrow X,$$

and

$$\frac{\varepsilon_{2n}}{\varepsilon_n} \left(\varepsilon_n \sum_{i=1}^n a_i \right) + \frac{\varepsilon_{2n}}{\varepsilon_n} \left(\varepsilon_n \sum_{i=n+1}^{2n} a_i \right) \Rightarrow X.$$

If we further suppose that there is $\alpha > 0$ such that

$$\lim_{n \rightarrow \infty} \frac{\varepsilon_{2n}}{\varepsilon_n} = \alpha,$$

we would have $\alpha X + \alpha \tilde{X} \sim X$, where $\tilde{X}$ is an independent copy of X . The only such distribution X with bounded second moment is the Gaussian distribution.

Recall that for independent random variables, we have a representation theorem that create new independent random variables on the product space. We also have a representation theorem for random variables converging in distribution. On the new probability space, they will converge a.e.

Theorem (Skorokhod's representation theorem). Suppose $X_n \Rightarrow X$ on some probability space (Ω, Σ, P) . Then there exists another probability space $(\tilde{\Omega}, \tilde{\Sigma}, \tilde{P})$ and random variables $\tilde{X}_n, \tilde{X}$ on it such that the following holds:

- $\tilde{X}_n \sim X_n$ for all n .
- $\tilde{X} \sim X$.
- $\tilde{X}_n \rightarrow \tilde{X}$ a.e.

Proof. Suppose for simplicity that F_{X_n} and F_X are all continuous and strictly increasing. With this, we can easily construct random variables with the same distribution. Let $U \sim \text{unif}([0, 1])$, then $F_X^{-1}(U) \sim X$. Indeed,

$$P(F_X^{-1}(U) \leq t) = P(F_X(F_X^{-1}(U)) \leq F_X(t)) = F_X(t).$$

Take $([0, 1], \mathcal{B}_{[0,1]}, m)$. Let $\tilde{X}_n = F_{X_n}^{-1}(\omega)$ and $\tilde{X} = F_X^{-1}(\omega)$. Then $\tilde{X}_n \sim X_n$ and $\tilde{X} \sim X$. Note that $X_n \Rightarrow X$ implies $F_{X_n} \rightarrow F_X$ pointwise. Checking $\tilde{X}_n \rightarrow \tilde{X}$ a.e. is left as an exercise. □

2.6 Law of rare events

As an example for convergence of random variables, we have the following example presenting the law of rare events. First we need some definitions.

Definition (Poisson distribution). Say random variable $X \sim \text{Poisson}(\lambda)$ has Poisson distribution with parameter λ for $\lambda > 0$ if

$$P(X = m) = \frac{\lambda^m e^{-\lambda}}{m!}.$$

Then, $EX = \lambda$, $EX^2 = \lambda^2 + \lambda$, and $\text{var } X = \lambda$.

Definition (exponential distribution). Say random variable $X \sim \text{Exp}(\lambda)$ has exponential distribution with parameter λ if X has density function

$$\rho_X(t) = \lambda \exp(-\lambda t) \quad \text{for } t \geq 0.$$

Then the corresponding CDF is

$$F_X(t) = \begin{cases} 1 - \exp(-\lambda t), & \text{if } t \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

The exponential distribution is a memoryless distribution: suppose $X \sim \text{Exp}(\lambda)$ and $t > s > 0$. Then,

$$P(X \geq t | X \geq s) = \frac{P(X \geq t)}{P(X \geq s)} = \exp(-\lambda(t - s)) = P(X \geq t - s).$$

Example (law of rare events). Let $\lambda > 0$ be a fixed parameter. Let $\{m_n\}_{n=1}^\infty$ be a non-decreasing sequence of integers such that

$$\lim_{n \rightarrow \infty} \frac{m_n}{n} = \lambda.$$

Create a triangular array as follows: for any n , let $\{U_{n,j}\}_{j=1}^{m_n}$ be i.i.d. uniform random variables on $[0, 1]$. Now let

$$X_n = |\{i \leq m_n : U_{n,i} \in [0, 1]\}|.$$

Then $X_n \Rightarrow \text{Poisson}(\lambda)$.

To prove this, let $k \geq 0$, we have

$$\begin{aligned} P(X_n = k) &= \binom{m_n}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{m_n - k} \\ &= (1 + o(1)) \frac{m_n^k}{k!} n^{-k} \exp\left(-\frac{m_n}{n}\right) \\ &\rightarrow (1 + o(1)) \frac{\lambda^k e^{-\lambda}}{k!} \end{aligned}$$

as $n \rightarrow \infty$. Therefore, $X_n \Rightarrow \text{Poisson}(\lambda)$ by the following proposition.

Proposition. Let $\{X_n\}_{n=1}^\infty$ and X be integer-valued random variables. Then $X_n \Rightarrow X$ iff $f_{X_n}(k) \rightarrow f_X(k)$ for any $k \in \mathbb{Z}$.

Proof. Verify directly using compactly supported continuous test functions. Since the test functions are compactly supported, only finite sums are involved.

□

Example. Continuing the previous example, define

$$Y_n = \min_{1 \leq i \leq m_n} U_{n,i}.$$

Then $Y_n \Rightarrow \text{Exp}(\lambda)$.

Let $t > 0$, then

$$\begin{aligned} P(Y_n \leq t) &= 1 - P(Y_n > t) \\ &= 1 - P\left(\bigcap_{i=1}^{m_n} U_{n,i} > t\right) \\ &= 1 - P(U_{n,1} > t)^{m_n} \\ &= 1 - \left(\frac{n-t}{n}\right)^{m_n} \\ &\rightarrow 1 - \exp(-\lambda t) \end{aligned}$$

as $n \rightarrow \infty$.

Example (Poisson process). Fix integer parameter $w > 0$. For every large n , let

$$Z_n = (U_{n,1}^*, U_{n,2}^*, \dots, U_{n,w}^*)$$

be w smallest numbers from the set $\{U_{n,i}\}_{i=1}^{m_n}$, arranged in increasing order, then Z_n is a random vector in $\mathbb{R}^w$. We want to investigate the limiting distribution of $\{Z_n\}_{n=1}^\infty$, if any exists.

Claim: let $\xi_1, \dots, \xi_w \sim \text{Exp}(\lambda)$ be i.i.d. Let the random vector

$$\xi = (\xi_1, \xi_1 + \xi_2, \dots, \xi_1 + \dots + \xi_w).$$

Then $Z_n \Rightarrow \xi$.

Example (Poisson process). Let $\{X_n\}_{n=1}^\infty$ be a sequence of i.i.d. random variables that follows the exponential distribution $\text{Exp}(\lambda)$. Define Poisson process $\{S_n\}_{n=0}^\infty$ via

$$S_n = \sum_{i=1}^n X_i.$$

This has two properties:

1. For each interval of length 1 on $\mathbb{R}_+$, the number of S_n 's within this interval follows the Poisson distribution with parameter λ .
2. For any finite collection of disjoint intervals, the variables corresponding to the number of S_n 's within the intervals are mutually independent. This is because of the memoryless property of the exponential distribution.

Lemma. If $c_j \rightarrow 0$, $a_j \rightarrow \infty$ and $a_j c_j \rightarrow \lambda$, then $(1 + c_j)^{a_j} \rightarrow e^\lambda$.

Theorem (Central limit theorem, basic form). Let $\{b_n\}_{n=1}^\infty$ be i.i.d. random variables that are Bernoulli($\frac{1}{2}$). Define $S_n = \sum_{i=1}^n b_i$. Then

$$\frac{S_n - \frac{n}{2}}{\sqrt{n}/2} \Rightarrow \mathcal{N}(0, 1).$$

Proof. Suppose n and k be some integer, then

$$P(S_{2n} = n + k) = \frac{(2n)!}{(n+k)!(n-k)!} 2^{-2n}$$

Using Sterling's formula $n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$, we have

$$\frac{(2n)!}{(n+k)!(n-k)!} \sim \frac{(2n)^{2n}}{(n+k)^{n+k}(n-k)^{n-k}} \frac{\sqrt{2\pi n}}{\sqrt{2\pi(n+k)}\sqrt{2\pi(n-k)}}$$

Hence,

$$P(S_n = n + k) \sim \left(1 + \frac{k}{n}\right)^{-n-k-\frac{1}{2}} \left(1 - \frac{k}{n}\right)^{-n+k-\frac{1}{2}} (\pi n)^{\frac{1}{2}}.$$

We also have

$$\left(1 + \frac{k}{n}\right)^{-n-k} \left(1 - \frac{k}{n}\right)^{-n+k} = \left(1 - \frac{k^2}{n^2}\right)^{-n} \left(1 + \frac{k}{n}\right)^{-k} \left(1 - \frac{k}{n}\right)^k.$$

Letting $k = x\sqrt{n/2}$, we know

$$\begin{aligned} \left(1 - \frac{k^2}{n^2}\right)^{-n} \left(1 + \frac{k}{n}\right)^{-k} \left(1 - \frac{k}{n}\right)^k &= \left(1 - \frac{x^2}{2n}\right)^{-n} \left(1 + \frac{x}{\sqrt{2n}}\right)^{-x\sqrt{n/2}} \left(1 - \frac{x}{\sqrt{2n}}\right)^{-x\sqrt{n/2}} \\ &\rightarrow \exp\left(\frac{x^2}{2}\right) \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{x^2}{2}\right) \\ &= \exp\left(-\frac{x^2}{2}\right) \end{aligned}$$

by the lemma above.

Therefore, if $2k/\sqrt{2n} \rightarrow x$, then

$$P(S_{2n} = n + k) \sim (\pi n)^{-1/2} e^{-x^2/2}.$$

Now,

$$\begin{aligned} P\left(a \leq \frac{S_{2n} - n}{\sqrt{n/2}} \leq b\right) &= P\left(n + a\sqrt{n/2} \leq S_{2n} \leq n + b\sqrt{n/2}\right) \\ &= \sum_{k \in [a\sqrt{n/2}, b\sqrt{n/2}] \cap \mathbb{Z}} P(S_{2n} = n + k). \end{aligned}$$

Chaging variables $x = k\sqrt{n/2}$, we then have

$$P\left(a \leq \frac{S_{2n} - n}{\sqrt{n/2}} \leq b\right) = \sum_{x \in [a, b] \cap (2\mathbb{Z}/\sqrt{2n})} (2\pi)^{-1/2} e^{-x^2/2} (2/n)^{1/2} \\ \approx \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx,$$

using the definition of Riemann integral. The proof is now complete. □

Definition (Multivariate Gaussian). There are two equivalent definitions of multivariate Gaussian.

1. $X \in \mathbb{R}^n$ has Gaussian distribution if there is non-random n -by- n non-random matrix and $\mu \in \mathbb{R}^n$ such that $X \sim MG + \mu$, where G is the standard Gaussian vector in $\mathbb{R}^n$. That is, $G = (G_1, \dots, G_n)$ are random vector with i.i.d. $\mathcal{N}(0, 1)$ components.
2. $X \in \mathbb{R}^n$ has Gaussian distribution if for all non-random $y \in \mathbb{R}^n$, $\langle X, y \rangle$ is a scalar Gaussian random variables.

Proof. *** TO-DO *** □

3 Law of large numbers

3.1 Weak law of large numbers

Theorem (Weak law of large numbers). Suppose $\{X_n\}_{n=1}^{\infty}$ are mutually independent, identically distributed random variables. Suppose also $E|X_1| < \infty$. Let $S_n = \sum_{i=1}^n X_i$, then $S_n/n \xrightarrow{p} EX_1$.

Proof. We prove this using truncation method. Let $\varepsilon > 0$ and choose truncation level $M(\varepsilon) > 0$ satisfying the following:

$$E|X_1 \mathbf{1}_{|X_1| > M}| < \varepsilon^2.$$

By dominated convergence theorem, there exists such an M . Note that this also implies

$$|EX_1 - E(X_1 \mathbf{1}_{|X_1| \leq M})| < \varepsilon.$$

For each n , we have

$$S_n = \sum_{k=1}^n X_k \mathbf{1}_{|X_k| \leq M} + \sum_{k=1}^n X_k \mathbf{1}_{|X_k| > M}.$$

Write

$$T_n = \sum_{k=1}^n X_k \mathbf{1}_{|X_k| \leq M} \quad \text{and} \quad R_n = \sum_{k=1}^n X_k \mathbf{1}_{|X_k| > M}.$$

For the first term, we can use Chebyshev's inequality to obtain

$$\begin{aligned} P\left(\left|\frac{T_n}{n} - E[X_1 \mathbf{1}_{|X_1| \leq M}]\right| \geq \varepsilon\right) &\leq \frac{1}{\varepsilon^2} \text{var}\left(\frac{T_n}{n}\right) \\ &= \frac{1}{n\varepsilon^2} \text{var}(X_1 \mathbf{1}_{|X_1| \leq M}) \\ &\leq \frac{4M^2}{n\varepsilon^2} \end{aligned}$$

Similarly, for the second term, we have

$$P\left(\left|\frac{R_n}{n}\right| \geq \varepsilon\right) \leq \frac{1}{\varepsilon} E\left|\frac{R_n}{n}\right| \leq \varepsilon.$$

It follows that

$$P\left(\left|\frac{S_n}{n} - EX_1\right| \geq 2\varepsilon + \varepsilon^2\right) \leq \frac{4M^2}{n\varepsilon^2} + \varepsilon.$$

□

Remark. Note that the only step we use independence is when we calculate $\text{var}(T_n/n)$. In fact, pairwise independence is sufficient for this step.

3.2 Borel-Cantelli lemmas

Theorem (Borel-Cantelli lemmas). Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of events and

$$P(A_n \text{ i.o.}) = P\left(\bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} A_n\right) = P(\limsup A_n)$$

1. If $\sum_{n=1}^{\infty} P(A_n) < \infty$, then $P(A_n \text{ i.o.}) = 0$.
2. Suppose also A_n 's are mutually independent and $\sum_{n=1}^{\infty} P(A_n) = \infty$, then $P(A_n \text{ i.o.}) = 1$.

Proof. 1. We have

$$P(A_n \text{ i.o.}) = P\left(\sum_{n=1}^{\infty} \mathbf{1}_{A_n} = \infty\right).$$

However, $E[\sum_{n=1}^{\infty} \mathbf{1}_{A_n}] = \sum_{n=1}^{\infty} P(A_n) < \infty$, so $P(A_n \text{ i.o.}) = 0$.

2. We have

$$P(A_n \text{ i.o.}) = P\left(\bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} A_n^c\right) \leq \sum_{k=1}^{\infty} P\left(\bigcap_{n=k}^{\infty} A_n^c\right).$$

Using independence and $1 - t \leq \exp(-t)$, we obtain

$$P\left(\bigcap_{n=k}^M A_n^c\right) = \prod_{n=k}^M (1 - P(A_n)) \leq \exp\left(-\sum_{n=k}^M P(A_n)\right) \rightarrow 0$$

as $M \rightarrow \infty$. Hence, $P(\bigcup_{n=k}^{\infty} A_n) = 1$ for all k . Since $\bigcup_{n=k}^{\infty} A_n \downarrow \limsup A_n$, it follows from monotone continuity that $P(A_n \text{ i.o.}) = 1$. □

Proposition. Let $\{b_n\}_{n=1}^{\infty}$ be mutually independent Bernoulli variables with parameters $\{p_n\}_{n=1}^{\infty}$ with $0 < p_n < 1$. Then, $b_n \rightarrow 0$ a.e. iff $\sum_{n=1}^{\infty} p_n < \infty$.

Proof. For any ω , $b_n(\omega) \rightarrow 0$ as $n \rightarrow \infty$ iff $b_n(\omega) = 1$ for at most finitely many n 's. Let $A_n = \{b_n = 1\}$. Then, $b_n \rightarrow 0$ a.e. iff for a.e. ω is contained in finitely many A_n 's. That is, $P(A_n \text{ i.o.}) = 0$. The proposition then follows from Borel-Cantelli lemma. □

3.3 Strong law of large numbers

Theorem. Let $\{X_n\}_{n=1}^\infty$ be mutually independent identically distributed random variables. Suppose $E|X_1| < \infty$ and let $S_n = \frac{1}{n} \sum_{k=1}^n X_k$. Then, $S_n \rightarrow EX_1$ a.e.

Proof. See notes for proof by Etemadi. □

Example (Coupon collector). Consider array of random variable $\{\{X_{j,n}\}_{j=1}^\infty\}_{n=1}^\infty$. For each n , the sequence $\{X_{j,n}\}_{j=1}^\infty$ are i.i.d. uniform on $[n] = \{1, \dots, n\}$. Let

$$T_n = \min \left\{ m \geq 1 : \bigcup_{i=1}^m \{X_{i,n}\} = [n] \right\}.$$

We want to show $T_n/(n \log n) \xrightarrow{p} 1$.

To show this, we use second moment method. For each $k \leq n$, let $T_{n,k}$ be the first time k coupons are collected. We then have $T_{n,0} = 0$ and

$$T_n = \sum_{k=1}^n (T_{n,k} - T_{n,k-1}).$$

Note that $T_{n,k} - T_{n,k-1}$ follows a geometric distribution. Hence,

$$\begin{aligned} E(T_{n,k} - T_{n,k-1}) &= \frac{n}{n-k+1}, \\ \text{var}(T_{n,k} - T_{n,k-1}) &= \frac{k-1}{n} \frac{1}{\left(1 - \frac{k-1}{n}\right)^2}. \end{aligned}$$

It follows that

$$ET_n = n \sum_{k=1}^n \frac{1}{n-k+1} \sim n \log n.$$

Also,

$$\text{var } T_n = \sum_{k=1}^n \text{var}(T_{n,k} - T_{n,k-1}) = \sum_{k=1}^n \frac{k-1}{n} \frac{1}{\left(1 - \frac{k-1}{n}\right)^2} = o(n^2 \log^2 n)$$

It then follows from Chebyshev's inequality that $T_n/(n \log n) \xrightarrow{p} 1$.

Example. Let $v_0 \in \{-1, 1\}^n$ and define a sequence of random vectors $\{X_i\}_{i=1}^\infty$ inductively as follows: define $X_0 = v_0$, and X_{i+1} is a random vector supported on $\{-1, 1\}^n$ obtained from X_i by selecting a random number $m \in [n]$ uniformly at random and flipping m -th component of X_i with probability $\frac{1}{2}$.

Now X_k is a random vector for each k . Denote the uniform distribution on $\{-1, 1\}^n$ as P , and we are interested in

$$\|X_k - P\|_{TV} = \inf_{\xi, \tilde{X}} P(\xi \neq \tilde{X}),$$

where the inf is taken over all pairs of random vectors ξ and $\tilde{X}$ such that $\tilde{X} \sim X_k$ and $\xi \sim P$.

Claim: to make sure that $\|X_k - P\|_{TV} = o(1)$, we must have $k = \Omega(n \log n)$.

This is similar to the coupon collector problem. We need to have selected at least $n(1 - O(n^{-1/2}))$ indices for the distribution of X_k to be roughly uniform. With a similar derivation to the coupon collector problem, this implies $k = \Omega(n \log n)$.

4 Central Limit theorem

4.1 Characteristic Function

Definition (characteristic function). Let X be a random vector in $\mathbb{R}^n$, we define the characteristic function (ch.f.) of $X : \mathbb{R}^n \rightarrow \mathbb{C}$ via

$$\varphi_X(t) = Ee^{i\langle X, t \rangle}$$

Suppose X has density ρ_X , then

$$\varphi_X(t) = \int_{\mathbb{R}^n} \exp(i\langle s, t \rangle) \rho_X(s) ds.$$

This is essentially the Fourier transform of ρ_X .

Proposition. Let X be a random variable, its characteristic function φ_X is uniformly continuous.

Proof. We have

$$|\varphi_X(t + \delta) - \varphi_X(t)| = \left| Ee^{i\langle X, t \rangle} (e^{i\langle X, \delta \rangle} - 1) \right| \leq E \left| e^{i\langle X, \delta \rangle} - 1 \right|.$$

Note that $|e^{i\langle X, \delta \rangle} - 1| \leq 2$. The proposition then immediately follows from dominated convergence theorem. $\square$

Proposition. If X is a random variable with $E|X|^k < \infty$ for some $k \geq 1$. Then φ_X is k times continuous differentiable and

$$\varphi^{(k)}(0) = i^k EX^k.$$

Proposition. Suppose X is random variable and $a, b \in \mathbb{R}$. Then

$$\varphi_{aX+b}(t) = \exp(itb)\varphi_X(at).$$

Proposition. If $X_1, X_2, \dots, X_n$ are mutually independent and $S = X_1 + \dots + X_n$, then

$$\varphi_S(t) = \prod_{k=1}^n \varphi_{X_k}(t).$$

Proposition. Suppose $X \sim \mathcal{N}(0, 1)$, then

$$\varphi_X(t) = \exp\left(-\frac{t^2}{2}\right).$$

Proof. Suppose $Y \sim \mathcal{N}(0, 1)$, and X, Y are independent. It follows that

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t) = \varphi_X(t)^2.$$

Meanwhile,

$$\varphi_{X+Y}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp(i(x+y)t) \exp\left(-\frac{x^2}{2}\right) \exp\left(-\frac{y^2}{2}\right) dx dy$$

Using change of variable with $u = x + y$, $v = x - y$, we obtain

$$\begin{aligned} \varphi_{X+Y}(t) &= \frac{1}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \exp\left(-\frac{1}{2}\left(\frac{u^2}{2} + \frac{v^2}{2}\right) + iut\right) du dv \\ &= \frac{1}{\sqrt{4\pi}} \int \exp\left(-\frac{1}{4}u^2 + itu\right) du \\ &= \varphi_X(\sqrt{2}t). \end{aligned}$$

Now we have a function equation

$$\varphi_X(\sqrt{2}t) = \varphi_X(t)^2.$$

This implies that $\varphi_X(t) = \exp(-t^2/2)$.

□

Now we present inversion formula, which shows the important correspondence between the ch.f. φ_X and the induced Borel measure μ_X of a random variable.

Theorem (inversion formula). Let X be a random variable. Let φ_X be the ch.f. and μ_X be the Borel measure on $\mathbb{R}$ induced by X . Suppose interval $(a, b]$ is such that $\mu_X(\{a, b\}) = 0$. Then,

$$\mu_X((a, b]) = \frac{1}{2\pi} \lim_{T \rightarrow \infty} \int_{-T}^T \left[\frac{\exp(-ita) - \exp(-itb)}{it} \varphi(t) \right] dt.$$

This shows that there is a unique correspondence between μ_X and φ_X .

Proof. Some application of Fubini-Tonelli theorem.

□

Remark. Using the formula, we can compute $\mu_X((a, b])$ in terms of φ_X even if $\mu_X(\{a, b\}) \neq 0$. Indeed, there are at most countably many x such that $\mu_X(\{x\}) \neq 0$, so there exists a sequence ε_m such that $\varepsilon_m \downarrow 0$, and $\mu_X(\{a + \varepsilon_m, b + \varepsilon_m\}) = 0$. Then

$$\mu_X((a, b]) = \lim_{m \rightarrow \infty} \mu_X((a_m, b_m]).$$

It is important that $\varepsilon_m \downarrow 0$ so that the approximation through monotone continuity is correct.

The same is true in the n -dimensional setting. Now

$$\varphi_X(t) = E \exp(i \langle t, X \rangle).$$

The induced Borel measure μ_X in $\mathbb{R}^n$ is uniquely determined by the ch.f. φ_X .

Theorem. If X has distribution density ρ_X , it can be expressed as

$$\rho_X(t) = \frac{1}{2\pi} \int_{\mathbb{R}} \exp(-its) \varphi_X(s) ds.$$

Theorem (Levy). Let $\{X_k\}_{k=1}^{\infty}$ be a sequence of random vectors in $\mathbb{R}^n$ and let $\{\varphi_k\}_{k=1}^{\infty}$ be the corresponding ch.f. Then the following holds

1. Suppose $X_k \Rightarrow X$ and let φ be the ch.f. of X . Then $\varphi_k(t) \rightarrow \varphi(t)$ as $k \rightarrow \infty$ for all $t \in \mathbb{R}^n$.
2. Suppose there exists φ that is continuous at zero and $\varphi_k(t) \rightarrow \varphi(t)$ for all $t \in \mathbb{R}^n$. Then, there exists X such that $\varphi_X = \varphi$ and $X_k \Rightarrow X$.

4.2 Central Limit Theorem

With the results above, we can prove the central limit theorem.

Theorem (central limit theorem). Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of i.i.d. random variable such that EX_1 and $\text{var } X_1$ are well-defined and finite. Let

$$S_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n (X_k - EX_k).$$

Then $S_n \Rightarrow \mathcal{N}(0, \text{var } X_1)$.

Proof. WLOG assume $EX_1 = 0$ and $\text{var } X_1 = 1$. Then

$$\varphi_{S_n}(t) = \prod_{k=1}^n \varphi_{\frac{X_k}{\sqrt{n}}}(t) = \left(\varphi_{X_1} \left(\frac{t}{\sqrt{n}} \right) \right)^n$$

Now, using Taylor's formula, we have

$$\varphi(\varepsilon) = 1 - \frac{\varepsilon^2}{2} + o(\varepsilon^2).$$

Therefore,

$$\varphi_{S_n}(t) = \left(1 + \frac{t^2}{2n} + o\left(\frac{t^2}{n}\right) \right)^n \rightarrow \exp\left(-\frac{t^2}{2}\right),$$

as $n \rightarrow \infty$ for any $t \in \mathbb{R}$. This completes the proof. □

Remark. Note that not only the partial sum S_n converges to normal in distribution, but the sup norm of the CDF of S_n and the CDF of normal distribution also converges to zero with a rate of $1/\sqrt{n}$. This is the optimal rate and is achieved by Bernoulli 1/2 variables.

Now we present another characterization of central limit theorem by Lindeberg. This characterization might provide more insight and intuition compared to the computational approach we used above with characteristic functions.

Definition (Lindeberg condition). Let $X_1, X_2, \dots$ be mutually independent random variables with zero mean. Suppose $\sigma_n^2 = \text{var } X_n < \infty$ and $b_n^2 = \sum_{i=1}^n \sigma_i^2$. Then X_i 's satisfy the *Lindeberg condition* if for any $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} b_n^{-2} \sum_{i=1}^n E \left[X_i^2 \mathbf{1}_{\{|X_i| > \varepsilon b_n\}} \right] = 0.$$

Theorem (Lindeberg CLT). Let $X_1, X_2, \dots$ be i.i.d. random variables that satisfy the Lindeberg condition. Define

$$S_n = \frac{1}{b_n} \sum_{i=1}^n X_i.$$

Then $S_n \Rightarrow \mathcal{N}(0, 1)$.

Proof. Use the definition of weak convergence in terms of test functions. WLOG, we can work with functions $h \in C_c^\infty(\mathbb{R})$, i.e. smooth and compactly supported functions. The goal is to show $Eh(S_n) \rightarrow Eh(\mathcal{N}(0, 1))$ as $n \rightarrow \infty$.

We need to use Taylor's formula:

$$h(x+u) = h(x) + uh'(x) + \frac{u^2}{2}h''(x) + O\left(\min\{|u|^2, |u|^3\}\right)$$

Also, write

$$S_n^{(j)} = \sum_{i=1}^j Y_i + \sum_{i=j+1}^n X_i,$$

where Y_i, X_i are mutually independent and $Y_i \sim \mathcal{N}(0, \sigma_i^2)$. Note that $b_n^{-1} S_n^{(n)} \sim \mathcal{N}(0, 1)$ and $S_n^{(0)} = \sum_{i=1}^n X_i$, so the goal now is to show

$$\left| Eh\left(b_n^{-1} S_n^{(n)}\right) - Eh\left(b_n^{-1} S_n^{(0)}\right) \right| = o(1).$$

It suffices to show that for all $1 \leq j \leq n$,

$$\left| Eh\left(b_n^{-1} S_n^{(j)}\right) - Eh\left(b_n^{-1} S_n^{(j-1)}\right) \right| = o(b_n^{-2} \sigma_j^2)$$

Now fix $1 \leq j \leq n$ and let

$$Z = \sum_{i=1}^{j-1} Y_i + \sum_{i=j+1}^n X_i.$$

Then, using Taylor's formula, we obtain

$$\begin{aligned} h(b_n^{-1} S_n^{(j)}) &= h(b_n^{-1} Z) + h'(b_n^{-1} Z) (b_n^{-1} S_n^{(j)} - b_n^{-1} Z) + \frac{1}{2} h''(b_n^{-1} Z) (b_n^{-1} S_n^{(j)} - b_n^{-1} Z)^2 \\ &\quad + O\left(\min\left\{b_n^{-2}(S_n^{(j)} - Z)^2, b_n^{-3}(S_n^{(j)} - Z)^3\right\}\right). \end{aligned}$$

Note that $S_n^{(j)} - Z = Y_j$ is independent of Z . It follows that

$$Eh(b_n^{-1} S_n^{(j)}) = Eh(b_n^{-1} Z) + \frac{1}{2} \frac{\sigma_j^2}{b_n^2} Eh''(b_n^{-1} Z) + O\left(E \min\left\{b_n^{-2} |Y_j|^2, b_n^{-3} |Y_j|^3\right\}\right)$$

Similarly,

$$Eh(b_n^{-1} S_n^{(j-1)}) = Eh(b_n^{-1} Z) + \frac{1}{2} \frac{\sigma_j^2}{b_n^2} Eh''(b_n^{-1} Z) + O\left(E \min\left\{b_n^{-2} |X_j|^2, b_n^{-3} |X_j|^3\right\}\right)$$

Hence,

$$\left| Eh(b_n^{-1} S_n^{(j)}) - Eh(b_n^{-1} S_n^{(j-1)}) \right| = O\left(E b_n^{-3} |Y_j|^3 + E \min\left\{b_n^{-2} |X_j|^2, b_n^{-3} |X_j|^3\right\}\right)$$

For simplicity, consider the classical setting where $\sigma_j^2 = 1$ for all j . Now $b_n = \sqrt{n}$. Note that the 3rd absolute moment of standard Gaussian is finite, we have

$$E b_n^{-3} |Y_j|^3 \leq O(b_n^{-3}) = O(n^{-3/2}) = o(n^{-1}).$$

For the second term, let $\varepsilon > 0$ be given and consider truncation:

$$\begin{aligned} E \min\left\{b_n^{-2} |X_j|^2, b_n^{-3} |X_j|^3\right\} &= E \min\left\{b_n^{-2} |X_j|^2, b_n^{-3} |X_j|^3\right\} (\mathbf{1}_{(|X_j| > M)} + \mathbf{1}_{(|X_j| \leq M)}) \\ &\leq E\left(b_n^{-2} |X_j|^2 \mathbf{1}_{(|X_j| > M)}\right) + E\left(b_n^{-3} |X_j|^3 \mathbf{1}_{(|X_j| \leq M)}\right) \end{aligned}$$

However,

$$E b_n^{-2} |X_j|^2 \mathbf{1}_{(|X_j| > M)} = \frac{1}{n} E (X_j \mathbf{1}_{|X_j| > M})^2 \rightarrow 0$$

as $M \rightarrow \infty$. We can then choose M such that

$$E(b_n^{-2} X_j^2 \mathbf{1}_{(|X_j| > M)}) \leq \frac{\varepsilon}{n}$$

Meanwhile,

$$E(b_n^{-3} |X_j|^3 \mathbf{1}_{(|X_j| \leq M)}) \leq \frac{M^3}{n^{3/2}}.$$

This shows that

$$E \min\left\{b_n^{-2} |X_j|^2, b_n^{-3} |X_j|^3\right\} = o(n^{-1})$$

and the proof is complete. $\square$

5 Martingales

5.1 Information and independence

Proposition. Suppose $\{\mathcal{F}_i\}_{i \in I}$ is a family of σ -fields, and $\mathcal{F}_i \subset \Sigma$ for some (Ω, Σ, P) . Let $\mathcal{E} \in \Sigma$. Suppose for any finite $J \subset I$, $\mathcal{E}$ is independent from $\sigma(\{\mathcal{F}_j : j \in J\})$. Then $\mathcal{E}$ is independent from $\sigma(\{\mathcal{F}_i : i \in I\})$.

Proof. Let Λ be the collection of events in Σ that are independent from $\mathcal{E}$. Then Λ is a λ -system from σ -additivity of P . Now let

$$\Pi = \{E \in \Sigma : E \in \sigma(\{\mathcal{F}_j : j \in J\}) \text{ for some } j \in J \text{ finite}\}.$$

Then, $\Pi \subset \Lambda$. Moreover, if $E_1, E_2 \in \Pi$ and let J_1, J_2 be the corresponding finite index set, then $E_1 \cap E_2 \in \sigma(\{\mathcal{F}_j : j \in J_1 \cup J_2\})$. Hence $E_1 \cap E_2 \in \Pi$ and Π is a π -system. Therefore, by π - λ theorem, we can conclude that $\Lambda \supset \sigma(\Pi) = \sigma(\{\mathcal{F}_i : i \in I\})$. □

Theorem (Kolmogorov's 0-1 law). Let (Ω, Σ, P) be probability space. Suppose $\{\mathcal{F}_i\}_{i=1}^\infty$ be a sequence of mutually independent σ -field of Σ . Define tail σ -field:

$$\mathcal{T} = \bigcap_{k=1}^{\infty} \sigma\left(\bigcup_{i=k}^{\infty} \mathcal{F}_i\right).$$

Then $P(E) \in \{0, 1\}$ for any $E \in \mathcal{T}$.

Proof. For any $k \geq 1$, we have $E \in \sigma(\bigcup_{i=k}^{\infty} \mathcal{F}_i)$, so E is independent from $\sigma(\mathcal{F}_1, \dots, \mathcal{F}_{k-1})$. The verification of the previous claim is left as an exercise. Now from the previous proposition, this implies that E is independent from $\sigma(\bigcup_{i=1}^{\infty} \mathcal{F}_i) \supset \mathcal{T}$. Therefore, E is independent from $\mathcal{T}$ and thus E is independent from itself. That is,

$$P(E) = P(E)^2,$$

and the desired results immediately follows. □

Exercise. Verify the claim in the proof above.

Proof. We just need to verify that if $\mathcal{F}_1, \dots, \mathcal{F}_n$ are independent, then $\mathcal{F}_1, \dots, \mathcal{F}_{k-1}, \sigma(\bigcup_{i=k}^n \mathcal{F}_i)$ are independent. Fix $E_1 \in \mathcal{F}_1, \dots, E_{k-1} \in \mathcal{F}_{k-1}$ and let Λ be the collection of sets $E \in \Sigma$ such that $P(E \cap E_1 \cap \dots \cap E_{k-1}) = P(E)P(E_1) \dots P(E_{k-1})$. It can be easily verified that this is a λ -system. Now let Π be the collection of sets in the form of $E_k \cap \dots \cap E_n$ where $E_k \in \mathcal{F}_k, \dots, E_n \in \mathcal{F}_n$. This is a π -system and generates $\sigma(\bigcup_{i=k}^n \mathcal{F}_i)$. Moreover, $\Pi \subset \Lambda$. The desired statement then follows from the π - λ system. □

Now we present a simple applications of the Kolmogorov's 0-1 law.

Proposition. Suppose $X_1, X_2, \dots$ are mutually independent random variables. Let $S_n = \sum_{i=1}^n X_i$ and $C = \{\lim_{n \rightarrow \infty} S_n/n \text{ exists}\}$. Then, $P(C) \in \{0, 1\}$.

Proof. Fix $k \geq 1$ and let $n \geq k$, then

$$\frac{S_n}{n} = \frac{X_1 + \dots + X_k}{n} + \frac{1}{n} \sum_{i=k+1}^n X_i.$$

The first term goes to zero as $n \rightarrow \infty$. Hence, $\lim_{n \rightarrow \infty} S_n/n$ exists iff $\lim_{n \rightarrow \infty} n^{-1} \sum_{i=k+1}^n X_i$ exists. This implies that $C \in \sigma(\bigcup_{i=k+1}^n X_i)$. Since this is true for all $k \geq 1$, C is in the tail σ -field. The proposition then follows from Kolmogorov's 0-1 law. □

5.2 Conditional expectation

The notion of conditional expectation is used to study independence between current element of a random process and its history.

As a elementary definition and construction, let (Ω, Σ, P) be a probability space. Suppose $\{A_j\}_{j \in J}$ be a finite partition of Ω into disjoint events that has positive probability. Suppose also X a random variable with $E|X| < \infty$. For each j , we can consider the conditional probability space and let E_j be the expectation of the restriction of X . Then,

$$E_j = \frac{1}{P(A_j)} E[X \mathbf{1}_{A_j}]$$

Now let Y be a new random variable on (Ω, Σ, P) defined via $Y(\omega) = E_j$ if $\omega \in A_j$. It follows that

$$EY = \sum_{j \in J} E_j P(A_j) = \sum_{j \in J} E[X \mathbf{1}_{A_j}] = EX.$$

Moreover, let $I \subset J$ and let $A_I = \bigcup_{j \in I} A_j$, then $\mathbf{1}_{A_I} = \sum_{j \in I} \mathbf{1}_{A_j}$. We can similarly obtain $E[Y \mathbf{1}_{A_I}] = E[X \mathbf{1}_{A_I}]$.

Now we present the formal definition.

Definition (Conditional expectation). Let (Ω, Σ, P) be a probability space. Suppose $\mathcal{F} \subset \Sigma$ be σ -field and random variable X with $E|X| < \infty$. Then the conditional expectation of X given $\mathcal{F}$ is a $\mathcal{F}$ -measurable random variable, write $E[X|\mathcal{F}]$ such that for any $A \in \mathcal{F}$,

$$\int_A X dP = \int_A E[X|\mathcal{F}] dP.$$

Remark. Note that if $\mathcal{F}$ is just $\{\emptyset, \Omega\}$, then the conditional expectation $E[X|\mathcal{F}] = EX$.

Theorem. Conditional expectation exists and is unique up to a null set.

Proof. (Uniqueness) Suppose both f and g are conditional expectation of X . Then

$$\int_A f dP = \int_A X dP = \int_A g dP$$

for all $A \in \mathcal{F}$. Hence, $f = g$.

(Existence) First suppose X is nonnegative. Define measure ν on $\mathcal{F}$ via

$$\nu(A) = \int_A X dP$$

for all $A \in \mathcal{F}$. This is a finite measure on $\mathcal{F}$, and it is absolutely continuous w.r.t. P . By Random-Nikodym theorem, there exists $\mathcal{F}$ -measurable function $f \geq 0$ such that $\nu(A) = \int_A f dP$. Therefore, f is a conditional expectation of X given $\mathcal{F}$. For real-valued X , let $f^\pm$ be the conditional expectation of $X^\pm$. Then, $f^+ - f^-$ is a conditional expectation for X . □

Proposition. We have the following properties of conditional expectation:

1. Suppose $E|X| < \infty$, $E|Y| < \infty$, and $X \leq Y$ a.e. Then, $E[X|\mathcal{F}] \leq E[Y|\mathcal{F}]$.
2. Suppose $X_n \uparrow X$ a.e., $X \geq 0$ and $E|X| < \infty$. Then, $E[X_n|\mathcal{F}] \uparrow E[X|\mathcal{F}]$ a.e.
3. If X is $\mathcal{F}$ -measurable, then $E[X|\mathcal{F}] = X$.
4. If X is independent of $\mathcal{F}$, then $E[X|\mathcal{F}] = EX$.
5. If $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}$, then $E[E[X|\mathcal{F}_2]|\mathcal{F}_1] = E[X|\mathcal{F}_1]$.

Example. Let $\{X_i\}_{i=1}^\infty$ be i.i.d. ± 1 symmetric variable, then $\sum_{i=1}^n X_i$ is the simple symmetric random walk on $\mathbb{Z}$. Define $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$, then $\{\mathcal{F}_n\}_{n=1}^\infty$ forms a filtration. We hope to compute the conditional expectation $E[S_n|\mathcal{F}_{n-1}]$.

Since $S_n = X_n + S_{n-1}$, we have

$$\begin{aligned} E[S_n|\mathcal{F}_{n-1}] &= E[S_{n-1}|\mathcal{F}_{n-1}] + E[X_n|\mathcal{F}_{n-1}] \\ &= S_{n-1} + EX_n \\ &= S_{n-1}. \end{aligned}$$

Remark. If $\{Y_n\}_{n=1}^\infty$ is a process such that $E(Y_n|Y_1, \dots, Y_{n-1}) = Y_{n-1}$, it is a discrete time martingale. In the example above $\{S_n\}_{n=1}^\infty$ is a discrete time martingale.

Remark. If $\{Y_n\}_{n=1}^\infty$ is on a countable state space and $F_{(Y_n|Y_1, \dots, Y_{n-1})} = F_{(Y_n|Y_{n-1})}$, we say $\{Y_n\}_{n=1}^\infty$ satisfies Markov property and is a Markov process. In the example above $\{S_n\}_{n=1}^\infty$ is a Markov process.

Definition (conditional probability). The conditional probability $P(A|\mathcal{F})$ is defined via

$$P(A|\mathcal{F}) = E[\mathbf{1}_A | \mathcal{F}].$$

Proposition. If X and Y are independent random variables, then $E[X|Y] = EX$.

Proof. For any event $A \in \sigma(Y)$, we have

$$\int_A E[X|Y] dP = \int_A X dP = \int X \mathbf{1}_A dP.$$

However,

$$\int X \mathbf{1}_A dP = E[X]E[\mathbf{1}_A] = P(A)E[X] = \int_A EX dP.$$

By uniqueness of conditional expectation, the proof is complete. □

Proposition. If X and Y are independent random variables and f is a Borel functions on $\mathbb{R}^2$, then

$$E[f(X, Y)|X] = g(X),$$

where $g(x) = Ef(x, Y)$ for all $x \in \mathbb{R}$.

Proof. Consider a setting where $X(\omega_1, \omega_2) = h(\omega_1)$, $Y(\omega_1, \omega_2) = z(\omega_2)$ and $\Omega = \Omega_X \times \Omega_Y$. Then for any event $A \in \Omega_X$, we have

$$\begin{aligned} \int_A g(X) &= \int_A \int_{\Omega_Y} f(X(\omega_1), Y(\omega_2)) dP(\omega_2) dP(\omega_1) \\ &= \int_{A \times \Omega_Y} f(X(\omega_1), Y(\omega_2)) dP(\omega_1, \omega_2). \end{aligned}$$

□

5.3 Reflection principle

Consider a random walk S_n on $\mathbb{Z}$. Let $\ell \leq 0$ and $u > \ell$ be integers and consider a n -time horizon. Start with a realization of $\{S_k\}_{k=1}^\infty$ such that $S_n = u$ and $S_k \leq k$ for some $k \leq n$. Then there is a bijection between paths that end at u and paths that end at $2\ell - u$. Let $M_n = \min_{0 \leq k \leq n} S_k$ be the running minimum of the random walk, then we have

$$P(S_n = u, M_n \leq \ell) = P(S_n = 2\ell - u).$$

Example. Now we can compute the distribution of M_n . Take $\ell \leq 0$, we have

$$\begin{aligned}
 P(M_n \leq \ell) &= P(M_n \leq \ell, S_n \leq \ell) + P(M_n \leq \ell, S_n > \ell) \\
 &= P(S_n \leq \ell) + P(M_n \leq \ell, S_n > \ell) \\
 &= P(S_n \leq \ell) + \sum_{u=\ell+1}^{\infty} P(M_n \leq \ell, S_n = u) \\
 &= P(S_n \leq \ell) + \sum_{u=\ell+1}^{\infty} P(S_n = 2\ell - u) \\
 &= 2P(S_n < \ell) + P(S_n = \ell).
 \end{aligned}$$

Therefore,

$$P(M_n = \ell) = P(S_n = \ell) + P(S_n = \ell - 1).$$

Then we can compute expectation:

$$\begin{aligned}
 EM_n &= \sum_{\ell=-\infty}^0 \ell P(M_n = \ell) \\
 &= \sum_{\ell=-\infty}^0 \ell P(S_n = \ell) + \sum_{\ell=-\infty}^{-1} (\ell + 1) P(S_n = \ell) \\
 &= \sum_{\ell=-\infty}^0 \ell P(S_n = \ell) + \sum_{\ell=1}^{\infty} (-\ell) P(S_n = \ell) + \sum_{\ell=-\infty}^{-1} P(S_n = \ell) \\
 &= E[-|S_n|] + P(S_n < 0).
 \end{aligned}$$

As $n \rightarrow \infty$, we have

$$\frac{EM_n}{E|S_n|} \rightarrow -1.$$

Hence,

$$EM_n = -\sqrt{2\pi^{-1}}\sqrt{n} + o(\sqrt{n})$$

Exercise. Use reflection principle to show that the distribution of the first return time $T = \min\{n > 0 : S_n = 0\}$ is

$$P(T = 2k) = \frac{1}{2k-1} 2^{-2k} \binom{2k}{k}.$$

Then show $P(T = \infty) = 0$. This is called the *recurrence* property.

Example (recurrence). From the previous exercise, we know the first return time T is finite with probability one. It follows that

$$P(S_n = 0 \text{ i.o.}) = 1.$$

Let T_1 be the time of first return to zero, i.e. $T_1 = \min\{t \geq 1 : S_t = 0\}$, and define a random sequence $\{S_{T_1+n}\}_{n=1}^\infty$. Let $i_1, \dots, i_k$ be fixed integers, we want to show

$$P(\cap_{j=1}^k S_{T_1+j} = i_j) = P(\cap_{j=1}^k S_j = i_j).$$

Towards using Markov property of random walk, we have

$$P(\cap_{j=1}^k S_{T_1+j} = i_j) = \sum_{\ell \geq 1} \sum_{u_1, \dots, u_{\ell-1}} P(T_1 = \ell, \cap_{j=1}^{\ell-1} S_j = u_j, \cap_{j=1}^k S_{\ell+j} = i_j)$$

For each term, we have

$$P(T_1 = \ell, \cap_{j=1}^{\ell-1} S_j = u_j, \cap_{j=1}^k S_{\ell+j} = i_j) = P(T_1 = \ell, \cap_{j=1}^{\ell-1} S_j = u_j) P(\cap_{j=1}^k S_{\ell+j} = i_j | T_1 = \ell)$$

The second term is the same for any $\ell \geq 0$, so the summation becomes

$$\begin{aligned} P(\cap_{j=1}^k S_{T_1+j} = i_j) &= \sum_{\ell \geq 1} P(T_1 = \ell) P(\cap_{j=1}^k S_{\ell+j} = i_j | T_1 = \ell) \\ &= P(\cap_{j=1}^k S_{T_1+j} = i_j) \end{aligned}$$

Therefore, $S_{T_1+n} \sim S_n$ are equidistributed.

Following this, we can also define T_k inductively as the time of k -th return to zero, and it is easy to see that $P(T_k < \infty \text{ for all } k) = 1$. Also, note that $\{T_k \leq \ell\} \in \sigma(S_1, \dots, S_\ell)$, this is called the *stopping time* property.

5.4 Subgaussian random variables

The covariance matrix of $X \in \mathbb{R}^n$ with $EX = 0$ is defined as

$$\text{cov}[X] = E[XX^T].$$

Note that $\text{tr } E[XX^T] = E[\text{tr } XX^T] = E[X^T X] = E\|X\|^2$.

Definition. Let X be a random vector in $\mathbb{R}^n$. Suppose $EX = 0$ and $\text{cov}[X] = \Sigma \in \mathbb{R}^{n \times n}$. If for any $y \in \mathbb{R}^n$ and $t > 0$,

$$P\left\{\langle X, y \rangle > t(E\langle X, y \rangle^2)^{1/2}\right\} \leq 2 \exp\left(-\frac{t^2}{k^2}\right)$$

We say X is a subgaussian vector.

If X is a subgaussian vector, then we have some constraints on the norm of X .

Theorem. Suppose X is a subgaussian random vector, then there exists $C_k, c_k > 0$ such that

$$P\left\{\|X\|_2 > t(E\|X\|_2^2)^{1/2}\right\} \leq C_k \exp(-c_k t^2)$$

for all $t > 0$.

Proof. The idea of the proof is to use test vectors and conditional expectation. Let $V \sim \mathcal{N}(0, I_n)$ and independent from X . Fix $t > 0$ and define event $A_t = \{\|X\| > t(E\|X\|^2)^{1/2}\}$. There exists some c such that for any $\tilde{x} \in \mathbb{R}^n$, $P\{|\langle V, \tilde{x} \rangle| \geq c\|\tilde{x}\|\} \geq \frac{3}{4}$. Define function f and g via

$$f(\tilde{x}, \tilde{v}) = \mathbf{1}_{\{|\langle \tilde{v}, \tilde{x} \rangle| \geq c\|\tilde{x}\|\}}, \quad g(\tilde{x}) = Ef(\tilde{x}, V) = P\{|\langle V, \tilde{x} \rangle| \geq c\|\tilde{x}\|\} \geq \frac{3}{4}.$$

Since V and X are independent, $E[f(X, V)|X] = g(X)$ a.e. It follows that

$$P\{|\langle V, X \rangle| \geq c\|X\| | X\} \geq \frac{3}{4} \text{ a.e.}$$

Next, since the components of V are i.i.d. $\mathcal{N}(0, 1)$, we have $E(V^T \Sigma V) = \text{tr } \Sigma = E\|X\|^2$. Note that Σ is PSD, Markov inequality gives

$$P\{V^T \Sigma V \leq 4E\|X\|^2\} = P\{V^T \Sigma V \leq 4E\|X\|^2 | X\} \geq \frac{3}{4}.$$

Combining the two inequalities, we obtain

$$P\{V^T \Sigma V \leq 4E\|X\|^2 \text{ and } |\langle V, X \rangle| \geq c\|X\| | X\} \geq \frac{1}{2} \text{ a.e.}$$

Define event $A = \{V^T \Sigma V \leq 4E\|X\|^2 \text{ and } |\langle V, X \rangle| \geq c\|X\|\}$.

Now we have $P(A \cap A_t) = P(A_t)P(A|A_t)$. Note that A_t is X -measurable and $P(A|X) \geq \frac{1}{2}$ a.e. Therefore, $P(A \cap A_t) \geq \frac{1}{2}P(A_t)$. Through computation we can obtain

$$A \cap A_t \subset \left\{ |\langle V, X \rangle| \geq \frac{1}{2}ct\sqrt{V^T \Sigma V} \right\}.$$

We now just need to bound the probability of the event on RHS.

For any $v \in \mathbb{R}^n$, define a new function h via

$$h(v) = E\left[\mathbf{1}_{\{|\langle v, X \rangle| \geq \frac{1}{2}ct\sqrt{v^T \Sigma v}\}}\right] = E\left[\mathbf{1}_{\{|\langle v, X \rangle| \geq \frac{1}{2}ct\sqrt{E\langle v, X \rangle^2}\}}\right]$$

Since X is subgaussian, $h(v) \leq 2 \exp\left(-\frac{c^2 t^2}{4k^2}\right)$ a.e. Therefore,

$$E\left[P\left\{|V, X| \geq \frac{1}{2}ct\sqrt{V^T \Sigma V} \middle| V\right\}\right] = Eh(V) \leq 2 \exp\left(-\frac{c^2 t^2}{4k^2}\right).$$

The LHS is exactly $P\{|\langle V, X \rangle| \geq \frac{1}{2}ct\sqrt{V^T \Sigma V}\}$, so

$$P(A_t) \leq 2P(A \cap A_t) \leq 4 \exp\left(-\frac{c^2 t^2}{4k^2}\right)$$

□

5.5 Martingales and concentration inequalities

Definition. Let $\{\mathcal{F}_n\}_{n=1}^\infty$ be a filtration and $\{X_n\}_{n=1}^\infty$ be an adapted sequence, i.e. X_n is $\mathcal{F}_n$ measurable. Suppose also $E|X_n| < \infty$ for all n . Then

1. $\{X_n\}_{n=1}^\infty$ is a submartingale if $E[X_{n+1}|\mathcal{F}_n] \geq X_n$ for all n .
2. $\{X_n\}_{n=1}^\infty$ is a supermartingale if $E[X_{n+1}|\mathcal{F}_n] \leq X_n$ for all n .
3. $\{X_n\}_{n=1}^\infty$ is a martingale if $E[X_{n+1}|\mathcal{F}_n] = X_n$ for all n .

Example. Suppose $\{Y_n\}_{n=1}^\infty$ are mutually independent nonnegative random variables with $EY_n = 1$. Define $X_n = \prod_{j=1}^n Y_j$. Then $\mathcal{F}_n = \sigma(Y_1, \dots, Y_n)$ is a filtration, and with this filtration X_n is a martingale.

Example. Suppose X is random variable with $E|X| < \infty$ and $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots$ is a filtration. Then $X_n = E[X|\mathcal{F}_n]$ is a martingale.

Next we are going to present some concentration inequalities based on martingales.

Theorem (Azuma's inequality). Let (Ω, Σ, P) be a probability space. Suppose $\{\mathcal{F}_i\}_{i=1}^N$ be a filtration with $N < \infty$. Suppose $\{X_i\}_{i=1}^N$ is a martingale and suppose the L^∞ norm $d_i = \|X_i - X_{i-1}\|_\infty < \infty$ for any $i \leq N$. Then

$$P\{|X_N - X_0| \geq t\} \leq 2 \exp\left(-\frac{t^2}{4 \sum_{i=1}^N d_i^2}\right)$$

for all $t > 0$.

Proof. We aim to estimate $P\{X_N - X_0 \geq t\}$. Fix $\lambda > 0$ and we have

$$\begin{aligned} E \exp(\lambda(X_N - X_0)) &= E \exp\left(\lambda \sum_{i=1}^N (X_i - X_{i-1})\right) \\ &= E \left[E \left[\exp\left(\lambda \sum_{i=1}^N (X_i - X_{i-1})\right) \middle| \mathcal{F}_{N-1} \right] \right] \\ &= E \left[\exp\left(\lambda \sum_{i=1}^N (X_i - X_{i-1})\right) E[\exp(\lambda(X_N - X_{N-1})) | \mathcal{F}_{N-1}] \right] \end{aligned}$$

No we need to use martingale property of X_n . However, we are dealing with the expectation of an exponential. Using $\exp x \leq x + \exp(x^2)$ for all $x \in \mathbb{R}$, we have

$$\begin{aligned} E[\exp(\lambda(X_N - X_{N-1})) | \mathcal{F}_{N-1}] &\leq E[\lambda(X_N - X_{N-1}) + \exp(\lambda^2(X_N - X_{N-1})^2) | \mathcal{F}_{N-1}] \\ &= E[\exp(\lambda^2(X_N - X_{N-1})^2) | \mathcal{F}_{N-1}] \\ &\leq \exp(\lambda^2 d_N^2). \end{aligned}$$

Therefore,

$$E \exp(\lambda(X_N - X_0)) \leq \exp(\lambda^2 d_N^2) E \left[\exp \left(\lambda \sum_{i=1}^N (X_i - X_{i-1}) \right) \right],$$

and inductively we can obtain $E \exp(\lambda(X_N - X_0)) \leq \exp \left(\lambda^2 \sum_{i=1}^N d_i^2 \right)$. Finally, Markov inequality gives

$$P \{ \exp(\lambda(X_N - X_0)) \geq \exp(\lambda t) \} \leq \frac{E \exp(\lambda(X_N - X_0))}{\exp(\lambda t)} \leq \exp \left(\lambda^2 \sum_{i=1}^N d_i^2 - \lambda t \right)$$

Letting $\lambda = \frac{1}{2} t (\sum_{i=1}^N d_i^2)^{-1}$ completes the proof. $\square$

Corollary (Bounded differences inequality). Let $\{\mathcal{F}_i\}_{i=1}^N$ be a filtration with $\mathcal{F}_N = \Sigma$ and $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Let ξ be a random variable with $E|\xi| < \infty$. Let $d_i = \|E[\xi|\mathcal{F}_i] - E[\xi|\mathcal{F}_{i-1}]\|_\infty$ for $1 \leq i \leq N$. Suppose that $d_i < \infty$ for all $i \leq N$, then

$$P \{ |\xi - E\xi| > t \} \leq 2 \exp \left(- \frac{t^2}{4 \sum_{i=1}^N d_i^2} \right)$$

for all $t > 0$.

Corollary. Suppose $Y_1, \dots, Y_N$ are mutually independent random variables. Suppose $f : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfies the following “Lipschitz” condition: for any $i \leq N$ and any choice of $x_j \in \text{range}(Y_i)$ for $j \neq i$ and $x_i, \tilde{x}_i \in \text{range}(Y_i)$,

$$|f(x_1, \dots, x_{i-1}, x_i, x_{i+1}, \dots, x_N) - f(x_1, \dots, x_{i-1}, \tilde{x}_i, x_{i+1}, \dots, x_N)| \leq 1.$$

Then

$$P \{ |f(Y_1, \dots, Y_N) - Ef(Y_1, \dots, Y_N)| > t \} \leq 2 \exp \left(- \frac{t^2}{4N} \right)$$

for any $t > 0$.

Proof. Consider a filtration given by $\mathcal{F}_0 = \{\emptyset, \Omega\}$, $\mathcal{F}_N = \Sigma$, and $\mathcal{F}_i = \sigma(Y_1, \dots, Y_i)$ for $1 \leq i < N$. Let $\xi = f(Y_1, \dots, Y_N)$ and consider

$$\begin{aligned} |E[\xi|\mathcal{F}_i] - E[\xi|\mathcal{F}_{i-1}]| &= |E_{Y_{i+1}, \dots, Y_N} f(Y_1, \dots, Y_N) - E_{Y_i, Y_{i+1}, \dots, Y_N} f(Y_1, \dots, Y_N)| \\ &= |E_{Y_{i+1}, \dots, Y_N} [f(Y_1, \dots, Y_N) - E_{Y_i} f(Y_1, \dots, Y_N)]| \\ &\leq E_{Y_{i+1}, \dots, Y_N} |f(Y_1, \dots, Y_N) - E_{Y_i} f(Y_1, \dots, Y_N)| \\ &\leq 1 \text{ a.e.} \end{aligned}$$

Therefore $d_i \leq 1$ and the desired statement follows from the previous theorem. $\square$

Example. Consider a discrete cube $\{-1, 1\}^n$ with the Hamming metric, i.e. the distance is the number of coordinates that does not agree. Suppose $f : \{-1, 1\}^n \rightarrow \mathbb{R}$ is Lipschitz with constant 1. Then, by previous theorems, we have

$$P\{|f(V) - Ef(V)| > t\} \leq 2 \exp\left(-\frac{t^2}{4n}\right)$$

for all $t > 0$, where V is the uniform random distribution on $\{-1, 1\}^n$.

Moreover, for $A \subset \{-1, 1\}^n$ with $|A| = 2^{n-1}$. We have the following *Hamming concentration inequality*:

$$P\{\text{dist}(X, A) \geq t\} \leq 2 \exp\left(-\frac{Ct^2}{n}\right)$$

for some universal constant C . From this we can obtain $E[\text{dist}(A, V)] = O(\sqrt{n})$.

Next we present a combinatorial application of martingale method.

Example. Let $G \sim G(n, \frac{1}{2})$ be a uniform distribution on labeled n -vertex graph. That is, the graph is constructed with $\binom{n}{2}$ independent edges $(b_{ij})_{i>j}$ following Bernoulli($\frac{1}{2}$) distribution. We want to consider the chromatic number $\chi(G)$ of the graph.

Define a filtration by $\mathcal{F}_m = \sigma(b_{ij} : j < i \leq m)$, $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Now the filtration $\mathcal{F}_0 \subset \mathcal{F}_1 \subset \dots \subset \mathcal{F}_n$ has n elements. Now $\{E[\chi(G)|\mathcal{F}_m]\}_{m=1}^n$ is a martingale, so by previous theorems,

$$P\{|\chi(G) - E\chi(G)| \geq t\} \leq 2 \exp\left(-\frac{t^2}{4 \sum_{i=1}^n d_i^2}\right),$$

where $d_i = \|E[\chi(G)|\mathcal{F}_i] - E[\chi(G)|\mathcal{F}_{i-1}]\|_\infty$. By intergrating out the extra variables (similar to previous example), we have

$$E[\chi(G)|\mathcal{F}_{l-1}] = E[E_{b_{lj}:l>j}[\chi(G)]|\mathcal{F}_l].$$

Hence,

$$d_l = \|E[\chi(G) - E_{b_{lj}:l>j}[\chi(G)]|\mathcal{F}_l]\|_\infty$$

Now for any graph Γ on $[n]$, let $T(\Gamma)$ be the set of all graphs $\tilde{\Gamma}$ on $[n]$ such that $\{i, j\}$, $i > j$ and $i \neq l$ is an edge of $\tilde{\Gamma}$ iff $\{i, j\}$ is an edge of Γ . That is, the graphs can only disagree on edges incident from l . Then we have

$$E_{b_{lj}:l>j}[\chi(G)] = \frac{1}{|T(G)|} \sum_{\tilde{\Gamma} \in T(G)} \chi(\tilde{\Gamma}).$$

This implies that $d_l \leq 1$ a.e., since disagreeing on one vertex change the chromatic by at most one. Thus $\text{var } \chi(G) = O(n)$ and $\chi(G) = E[\chi(G)](1 \pm o(1))$. Note $E[\chi(G)] \sim \frac{n}{\log n}$.

5.6 Predicatble sequences and stopping time

Definition (predicatble sequence). Let $\{\mathcal{F}_n\}_{n=0}$ be a finite or infinite filtration and $\{H_n\}_n$ be a sequence of random variables. Say $\{H_n\}$ is predicatble if for any n , H_n is $\mathcal{F}_{n-1}$ measurable.

Now let X_n be an adapted sequence, and define a sequence $\{H \cdot X\}_n$ via

$$(H \cdot X)_n = \sum_{i=1}^n H_i(X_i - X_{i-1})$$

for $n \geq 0$.

Theorem. Suppose $\{H_n\}$ is non-negative predicatble sequence and $\{X_n\}$ is a submartingale (resp. supermartingale). Then $\{H \cdot X\}_n$ is a submartingale (resp. supermartingale).

Proof. Note that

$$\begin{aligned} E \left[\sum_{i=1}^n H_i(X_i - X_{i-1}) \middle| \mathcal{F}_{n-1} \right] &= \sum_{i=1}^{n-1} H_i(X_i - X_{i-1}) + E[H_n(X_n - X_{n-1}) | \mathcal{F}_{n-1}] \\ &= \sum_{i=1}^{n-1} H_i(X_i - X_{i-1}) + H_n E[X_n - X_{n-1} | \mathcal{F}_{n-1}]. \end{aligned}$$

The statement then follows immediately. $\square$

The main applications of predicatble sequences are related to stopping times.

Definition (stopping time). A random variable T is a stopping time w.r.t. filtration $\{\mathcal{F}_n\}_{n=1}^\infty$ if for any $k \in \mathbb{N}$, the event $\{T \leq k\}$ is $\mathcal{F}_k$ -measurable.

Example. Let $\{X_n\}_{n=0}^\infty$ be a martingale w.r.t filtration $\{\mathcal{F}_n\}_{n=0}^\infty$ with $\mathcal{F}_0$ being the trivial σ -field. Let T be a stopping time w.r.t. $\{\mathcal{F}_n\}_{n=0}^\infty$ such that there exists $M < \infty$ satisfying $P\{T \leq M\} = 1$. Then $EX_T = X_0$.

We can write

$$X_T = X_0 + \sum_{i=1}^M \mathbf{1}_{\{T \geq i\}}(X_i - X_{i-1}).$$

Now for $i \geq 0$, define $H_i = \mathbf{1}_{\{T \geq i\}} = 1 - \mathbf{1}_{\{T \leq i-1\}}$, so H_i is a predicatble sequence. By the previous theorem, the statement follows.

Next we consider the important optional stopping time theorem.

Theorem (Optional stopping time theorem). Let $\{\mathcal{F}_n\}_{n=0}^\infty$ be a filtration, T be a stopping time, and $\{X_n\}_{n=0}^\infty$ an adapted sequence. Then if $\{X_n\}_{n=0}^\infty$ is a submartingale (resp. supermartingale), then $\{X_{n \wedge T}\}_{n=0}^\infty$ is also a submartingale (resp. supermartingale) w.r.t the same filtration $\{\mathcal{F}_n\}_{n=0}^\infty$. Moreover, for any fixed $M \geq 0$, the sequence $\{X_{(n \vee T) \wedge M}\}_{n=0}^\infty$ is a submartingale (resp. supermartingale) w.r.t the filtration $\{\sigma(X_{(j \vee T) \wedge M}, 0 \leq j \leq m)\}_{m=0}^\infty$.

Proof. For the first part, we have

$$X_{T \wedge n} - X_0 = \sum_{k=1}^n \mathbf{1}_{\{T \geq k\}} (X_k - X_{k-1}).$$

Since $\mathbf{1}_{T \leq k}$ is a predictable sequence, $X_{T \wedge n}$ is a submartingale.

For the second part, write $\mathcal{G}_m = \sigma(X_{(j \vee T) \wedge M}, 0 \leq j \leq m)$. Fix any $A \in \mathcal{G}_{n-1}$ with $P(A) > 0$ and any $k \in \mathbb{N} \cup \{\infty\}$ such that $P(A \cap \{T = k\}) > 0$. The event A is representable as $\{(X_{M \wedge (m \vee T)})_{m=0}^{n-1} \in B\}$ for some $B \in \mathcal{B}(\mathbb{R}^n)$. Next consider

$$E \left[(X_{M \wedge (T \vee n)} - X_{M \wedge (T \vee (n-1))}) \mathbf{1}_{A \cap \{T=k\}} \right].$$

There are several cases.

If $k \geq n$, then $T \vee n = T \vee (n-1) = T = k$ on $\{T = k\}$, and the conditional expectation is zero.

If $k \leq n-1$, then $T \vee n = n$ and $T \vee (n-1) = n-1$. The conditional expectation becomes

$$E \left[(X_{M \wedge n} - X_{M \wedge (n-1)}) \mathbf{1}_{A \cap \{T=k\}} \right].$$

If $M \leq n-1$, then this becomes zero too. Therefore the only nontrivial case is when $M \geq n$. For this case, the conditional expectation becomes $E[X_n - X_{n-1} \mid A \cap \{T = k\}]$. Note that on the event $\{T = k\}$ with $k \leq n-1$, for all $1 \leq j \leq n-1$ we have

$$(j \vee T) \wedge M \leq (j \vee (n-1)) \wedge M \leq n-1.$$

Therefore, A is $\mathcal{F}_{n-1}$ -measurable. Also, T is a stopping time, so $\{T = k\}$ is also $\mathcal{F}_{n-1}$ -measurable. Now submartingale property implies

$$E[(X_n - X_{n-1}) \mathbf{1}_{A \cap \{T=k\}}] = E[\mathbf{1}_{A \cap \{T=k\}} E[X_n - X_{n-1} \mid \mathcal{F}_{n-1}]] \geq 0.$$

Therefore,

$$E \left[(X_{M \wedge (T \vee n)} - X_{M \wedge (T \vee (n-1))}) \mathbf{1}_{A \cap \{T=k\}} \right] \geq 0.$$

Now since $A = \bigsqcup_{k \in \mathbb{N} \cup \{\infty\}} A \cap \{T = k\}$, we have

$$\begin{aligned} E \left[(X_{M \wedge (T \vee n)} - X_{M \wedge (T \vee (n-1))}) \mathbf{1}_A \right] &= \sum_{k \in \mathbb{N} \cup \{\infty\}} E \left[\mathbf{1}_{A \cap \{T=k\}} (X_{M \wedge (T \vee n)} - X_{M \wedge (T \vee (n-1))}) \right] \\ &\geq 0. \end{aligned}$$

for any $A \in \mathcal{G}_{n-1}$. This then implies that

$$E \left[X_{M \wedge (T \vee n)} - X_{M \wedge (T \vee (n-1))} \mid \mathcal{G}_{n-1} \right] \geq 0,$$

as desired. □

Theorem (maximal inequalities). Let $\{X_n\}_{n=0}^\infty$ be a submartingale. Then,

$$aP\left\{\max_{0 \leq i \leq n} X_i > a\right\} \leq E[X_n \mathbf{1}_{\{\max_{0 \leq i \leq n} X_i > a\}}] \leq E|X_n|$$

for any $a > 0$.

Proof. Let stopping time $T = \inf\{m \geq 0 : X_m > a\}$. Then,

$$\begin{aligned} aP\left\{\max_{0 \leq i \leq n} X_i > a\right\} &\leq E[\mathbf{1}_{\{\max_{0 \leq i \leq n} X_i > a\}} X_{T \wedge n}] \\ &= EX_{T \wedge n} - E[\mathbf{1}_{\{\max_{0 \leq i \leq n} X_i \leq a\}} X_{T \wedge n}] \\ &= EX_{T \wedge n} - E[\mathbf{1}_{\{\max_{0 \leq i \leq n} X_i \leq a\}} X_n] \end{aligned}$$

By optional stopping time theorem, $\{X_{n \wedge (T \vee m)}\}_{m=1}^\infty$ is a submartingale. Then the expectation of the n -th element is at least the expectation of the 0-th element. This implies $EX_{T \wedge n} \leq EX_n$. Alternatively, we can also see this from

$$X_n - X_{n \wedge T} = \sum_{i=1}^n \mathbf{1}_{\{T < i\}} (X_i - X_{i-1}).$$

Rearranging the inequality gives the desired result. □

Corollary (Kolmogorov's maximal inequality). Let $\{X_n\}_{n=1}^\infty$ be mutually independent, $EX_n = 0$, and $\text{var } X_n < \infty$. Then,

$$P\left\{\max_{1 \leq n \leq m} \left|\sum_{i=1}^n X_i\right| > a\right\} \leq \frac{1}{a^2} \sum_{n=1}^m \text{var } X_n$$

for any $a > 0$.

Proof. Let $Y_n = (\sum_{i=1}^n X_i)^2$ for $n \geq 1$. We can verify that Y_n is a submartingale. It then follows from the previous theorem that

$$P\left\{\max_{1 \leq n \leq m} Y_n > a^2\right\} \leq \frac{1}{a^2} E|Y_m| = \frac{1}{a^2} \sum_{n=1}^m \text{var } X_n,$$

as desired. □

Without martingale theory, we can also provide a more elementary proof for Kolmogorov's inequality for symmetric variables.

Theorem. Let $\{X_n\}_{n=1}^\infty$ be mutually independent and symmetric random variables. Suppose also $EX_n = 0$, and $\text{var } X_n < \infty$. Then,

$$P \left\{ \max_{1 \leq n \leq m} \sum_{i=1}^n X_i > a \right\} \leq \frac{1}{a^2} \sum_{n=1}^m \text{var } X_n$$

for any $a > 0$.

Proof. Write $S_n = \sum_{i=1}^n X_i$. We have

$$\begin{aligned} P \left\{ \max_{1 \leq i \leq m} S_i > a \right\} &\leq P \{S_m > a\} + P \{\exists i < m : S_i > a \text{ and } S_m < S_i\} \\ &= P \{S_m > a\} + \sum_{i=1}^{m-1} P \{S_i > a, S_r \leq a \text{ for all } r < i, \text{ and } S_m - S_i < 0\} \\ &= P \{S_m > a\} + \sum_{i=1}^{m-1} P \{S_i > a, S_r \leq a \text{ for all } r < i, \text{ and } S_m - S_i > 0\}, \end{aligned}$$

where in the last step we used the fact that X_i 's are symmetric. Then, combining $S_i > a$ and $S_m - S_i > 0$, we can obtain

$$P \{\max S_i > a\} \leq 2P \{S_m > a\} = P \{|S_m| > a\} \leq \frac{1}{a^2} \sum_{i=1}^m \text{var } X_i.$$

□

5.7 Martingale convergence

Theorem (upcrossing lemma). Let $\{X_n\}_{n=0}^\infty$ be a submartingale and fix $a < b$. Next define stopping times $\{T_k\}_{k=0}^\infty$ taking values in $\mathbb{N} \cup \{\infty, 0, -1\}$ via $T_0 = -1$, and for $k \geq 1$,

$$T_{2k-1} = \inf\{l > T_{2k-2} : X_l \leq a\}, \quad T_{2k} = \inf\{l > T_{2k-1} : X_l \geq b\}.$$

Also, for all $m \geq 0$, let $U_m = \max\{k \geq 0 : T_{2k} \leq m\}$. Then, for any $m \geq 0$,

$$(b-a)E[U_m] \leq E[(X_m - a)^+] - E[(X_0 - a)^+].$$

Remark. If X_n is a submartingale, then $(X_n - a)^+$ is also a submartingale. Indeed, since $\varphi(x) = \max(x, 0)$ is convex and monotone,

$$E[\varphi(X_{m+1})|\mathcal{F}_m] \geq \varphi(E[X_{m+1}|\mathcal{F}_m]) \geq \varphi(X_m).$$

Proof. For $n \geq 1$, define

$$H_n = \mathbf{1}_{\{T_{2k-1} < n \leq T_{2k} \text{ for some } k \geq 1\}}.$$

Claim that H_n is predicatble. Indeed, note that $H_n = 1$ iff $\max\{l : T_l < n\}$ is odd, and this is $\mathcal{F}_{n-1}$ measurable. Write $Z_n = (X_n - a)^+$. Claim that

$$(b-a)U_m \leq \sum_{i=1}^{\max\{T_{2l}: T_{2l} \leq m\}} H_i(Z_i - Z_{i-1}) = (H \cdot Z)_{\max\{T_{2l}: T_{2l} \leq m\}}.$$

Indeed, each upcrossing contributes at least $b-a$ to the summation. By the condition of $H_n = 1$ above, we can verify that

$$\sum_{i=\max\{T_{2l}: T_{2l} \leq m\}+1}^m H_i(Z_i - Z_{i-1}) \geq 0.$$

Hence, $(b-a)U_m \geq (H \cdot Z)_m$. Now note that Z is a submartingale and $1-H$ is predicatble, so $((1-H) \cdot Z)$ is also a submartingale. Therefore,

$$\begin{aligned} (b-a)E[U_m] &\leq E(H \cdot Z)_m \\ &\leq E(H \cdot Z)_m + E((1-H) \cdot Z)_m \\ &= E(1 \cdot Z)_m \\ &= EZ_m - EZ_0. \end{aligned}$$

□

Now an application of the upcrossing lemma.

Example. Suppose $E[(X_n - a)^+]$ is uniformly bounded, i.e.

$$\sup_n E[(X_n - a)^+] < \infty.$$

This implies that EU_m is also uniformly bounded.

Consider the event A that there is infinitely many oscillations. Then everywhere on A , we have $\lim_{m \rightarrow \infty} U_m = \infty$. Hence if $P(A) > 0$, then $EU_m \rightarrow \infty$ as $m \rightarrow \infty$, a contradiction.

We can also use this to consider a symmetric random walk on the real numbers.

Example. Fix $0 < \gamma < 1$. Let $\{Y_n\}_{n=1}^\infty$ be mutually independent and symmetric random variable with

$$P\{Y_n = n^{-\gamma}\} = P\{Y_n = -n^{-\gamma}\} = \frac{1}{2}.$$

Let $X_n = \sum_{j=1}^n Y_j$. Then X_n is a martingale and we hope to investigate the trajectory of X_n .

We have

$$\text{var } X_n = \sum_{k=1}^n \frac{1}{k^{2\gamma}}.$$

Hence, if $\gamma > \frac{1}{2}$, the variance of X_n are uniformly bounded. Therefore, for any $a \in \mathbb{R}$, $E[(X_n - a)^+]$ is also uniformly bounded, and thus by our previous example, for any $a < b$, there are finitely many oscillations with probability 1. From the previous example and some extra argument, we can derive that the random walk converges to a limit that is nonzero. Indeed, the variance is nonzero, so the limit is not a.s. equal to zero. From Kolmogorov 0-1 law, the limit is nonzero a.s.

Theorem (submartingale convergence theorem). Let $\{X_n\}_{n=1}^\infty$ is a submartingale with

$$\sup_n E[X_n^+] < \infty.$$

Then there exists random variable X such that $E|X| < \infty$ and $X_n \rightarrow X$ a.e.

Proof. Let $S = \limsup_n X_n$ and $L = \liminf_n X_n$. Then we have $L^+ \leq \liminf_n X_n^+$ and it follows from Fatou's lemma that

$$E[L^+] \leq E[\liminf_n X_n^+] \leq \liminf_n E[X_n^+] < \infty$$

Also,

$$\sup_n E[(-X_n)^+] = \sup_n (E[X_n^+] - EX_n) \leq \sup_n E[X_n^+] - EX_1 < \infty.$$

Now $-S = \liminf_n (-X_n)$ so it follows similarly from Fatou's lemma that $E[(-S)^+] < \infty$. Therefore, $S \neq \infty$ and $L \neq -\infty$ a.e.

Now suppose $P\{S - L > \varepsilon\} > 0$. Then there exists $a < b$ such that $P\{L < a < b < S\} > 0$. This implies that there is a positive probability that infinitely oscillations happens between a, b and thus $EU_m \rightarrow \infty$ as $m \rightarrow \infty$, a contradiction with $\sup_n E[X_n^+] < \infty$. Moreover, $E|X| < \infty$ since $E[X^+] = E[L^+] < \infty$ and $E[X^-] = E[(-S)^+] < \infty$.

□

Corollary. Let $\{X_n\}_{n=1}^\infty$ be a non-negative supermartingale. Then there exists random variable X such that $EX \leq EX_n$ for any $n \geq 0$ and $X_n \rightarrow X$ a.e.

Proof. Note that $-X_n$ is a submartingale and $-X_n \leq 0$. Hence $\sup_n E[(-X_n)^+] = 0$, so existence of X is guaranteed by the previous theorem. Moreover,

$$E[X] \leq \liminf_{n \rightarrow \infty} E[X_n] = \inf_n E[X_n]$$

since X_n is a supermartingale, completing the proof.

□

Example. Let $\xi_1, \xi_2, \dots$ i.i.d. non-negative random variables with $E\xi_1 \leq 1$. Then $X_n = \prod_{i=1}^n \xi_i$ is a supermartingale, and $X_n \rightarrow 0$ a.e. unless $\xi_1 = 1$ a.e.

To prove this, suppose $\xi_1 \neq 1$ then there exists $\varepsilon > 0$ such that $P\{|\xi_1 - 1| > \varepsilon\} > \varepsilon$. Then, the event $A = \{|\xi_i - 1| \geq \varepsilon \text{ i.o.}\}$ has probability one. On event A ,

$$|X_{m+1} - X_m| = \left| \prod_{i=1}^{m+1} \xi_i - \prod_{i=1}^m \xi_i \right| \geq \varepsilon \prod_{i=1}^m \xi_i$$

for all m such that $|\xi_{m+1} - 1| \geq \varepsilon$, and i.e. for infinitely many indices. By convergence theorem, there exists random variable X such that $X_n \rightarrow X$ a.e. Let B be the event $X_n \rightarrow X$.

To show $X = 0$, suppose for contradiction that there exist $\delta > 0$ such that $P\{X > \delta\} > \delta$, then let event $C = \{X > \delta\}$. Consider the intersection $A \cap B \cap C$. Let $\omega \in A \cap B \cap C$. Then for all sufficient large n , $X_n(\omega) > \delta$. Then for infinitely many m 's we have $|X_{m+1}(\omega) - X_m(\omega)| \geq \varepsilon \delta$. Therefore, $\lim_{n \rightarrow \infty} X_n(\omega)$ does not exist, a contradiction.

Example (discrete-time birth process). Let $\{\xi_{g,i}\}_{g,i=1}^\infty$ be i.i.d. non-negative integer valued random variables. Define $\{X_n\}_{n=1}^\infty$ inductively by $X_0 = 1$ and $X_n = \sum_{i=1}^{X_{n-1}} \xi_{n,i}$ for $n \geq 1$. We hope to investigate $P\{x_n > 0 \text{ for all } n \geq 1\}$. Claim that

- $E\xi_{1,1} > 1$ implies $P\{x_n > 0 \text{ for all } n \geq 1\} > 0$.
- $E\xi_{1,1} < 1$ implies $P\{x_n > 0 \text{ for all } n \geq 1\} = 0$.
- $E\xi_{1,1} = 1$ and $\xi_{1,1} \neq 1$ implies $P\{x_n > 0 \text{ for all } n \geq 1\} = 0$.

First consider the case where $E\xi_{1,1} < 1$. Claim that X_n is non-negative supermartingale. Indeed,

$$\begin{aligned} E[X_n | \mathcal{F}_{n-1}] &= E \left[\sum_{i=1}^{X_{n-1}} \xi_{n,i} \middle| \mathcal{F}_{n-1} \right] \\ &= E \left[\sum_{i=1}^{\infty} \xi_{n,i} \mathbf{1}_{\{i \leq X_{n-1}\}} \middle| \mathcal{F}_{n-1} \right] \\ &= \sum_{i=1}^{\infty} \mathbf{1}_{\{i \leq X_{n-1}\}} E\xi_{1,1} \\ &\leq X_{n-1}. \end{aligned}$$

Then there exists non-negative X such that $X_n \rightarrow X$ a.e. Moreover, it is easy to verify that $\lim_{n \rightarrow \infty} EX_n = 0$. Hence by Fatou's lemma, $0 = \lim_{n \rightarrow \infty} EX_n \geq E \lim_{n \rightarrow \infty} X_n$, which implies $EX = 0$ and thus $X = 0$ a.e.

Next consider the case where $E\xi_{1,1} = 1$ with $\xi_{1,1}$ non-constant. We still have that X_n is a non-negative supermartingale. Therefore, there exists $X \geq 0$ such that $X_n \rightarrow X$ a.e. It is easy to see that X is integer-valued. Suppose there exists $m \geq 1$ such that $P\{X = m\} > 0$. On the event $\{X = m\}$, $X_n = m$ for all but finitely many n 's. However, since $\xi_{1,1} \neq 1$, X_n does not a.s.

equals m given $X_{n-1} = m$. Therefore, for some $\delta > 0$,

$$\begin{aligned} P\{X_n = m \text{ for all } n_0 \leq n \leq N\} &= \prod_{n=n_0+1}^N P\{X_n = m | X_j = m \text{ for all } n_0 \leq j \leq n\} \\ &\leq (1 - \delta)^{N-n_0}, \end{aligned}$$

a contradiction.

Finally, consider the last case where $E\xi_{1,1} > 1$. We first introduce the concept of *stochastic domination*. We say random variable Z *stochastically dominates* W if $P\{W \leq t\} \geq P\{Z \leq t\}$ for all $t \in \mathbb{R}$. In this case, there exists $\tilde{W} \sim W$ and $\tilde{Z} \sim Z$ on the same probability space such that $\tilde{W} \leq \tilde{Z}$ a.e.

Let $M > 0$ and $\varepsilon > 0$ be such that $\tilde{\xi}_{1,1} = \min(\xi_{1,1}, M)$ satisfy $E\tilde{\xi}_{1,1} \geq 1 + \varepsilon$. Also define an auxiliary process via $\tilde{X}_0 = 1$ and $\tilde{X}_n = \sum_{i=1}^{\tilde{X}_{n-1}} \tilde{\xi}_{n,i}$. We can verify that X_n stochastically dominates $\tilde{X}_n$ and $\tilde{X}_n \leq X_n$ a.e. for any n .

Claim that $P\{\tilde{X}_n > 0 \text{ for all } n \geq 1\} > 0$. Write $\mathcal{F}_n = \sigma(\tilde{X}_1, \dots, \tilde{X}_n)$. Since $E\tilde{\xi}_{1,1} \geq 1 + \varepsilon$, we have $E[\tilde{X}_n | \tilde{X}_{n-1}] \geq (1 + \varepsilon)\tilde{X}_{n-1}$ a.e. Hence,

$$\begin{aligned} P\left\{\tilde{X}_n \leq \left(1 + \frac{\varepsilon}{2}\tilde{X}_{n-1}\right) \middle| \mathcal{F}_{n-1}\right\} &\leq P\left\{\tilde{X}_n \leq \frac{1 + \frac{\varepsilon}{2}}{1 + \varepsilon} E[\tilde{X}_n | \tilde{X}_{n-1}] \middle| \mathcal{F}_{n-1}\right\} \\ &\leq P\left\{\tilde{X}_n \leq \left(1 - \frac{\varepsilon}{4}\right) E[\tilde{X}_n | \tilde{X}_{n-1}] \middle| \mathcal{F}_{n-1}\right\}, \end{aligned}$$

where in the last step we chose $\varepsilon > 0$ to be small enough such that

$$\frac{1 + \frac{\varepsilon}{2}}{1 + \varepsilon} \leq 1 - \frac{\varepsilon}{4}.$$

Write $A_k = \{\tilde{X}_{n-1} = k\}$ with $P(A_k) > 0$, then for $k > 0$ we have

$$P\left\{\tilde{X}_n \leq \left(1 - \frac{\varepsilon}{4}\right) E[\tilde{X}_n | \tilde{X}_{n-1} = k] \middle| \tilde{X}_{n-1} = k\right\} \leq \frac{\text{var}[\tilde{X}_n | \tilde{X}_{n-1} = k]}{(\frac{\varepsilon}{4})^2 E[\tilde{X}_n | \tilde{X}_{n-1} = k]^2}.$$

For $k = 0$ simply bound the probability with 1. Now,

$$\text{var}[\tilde{X}_n | \tilde{X}_{n-1} = k] = k \text{var} \tilde{\xi}_{1,1} \leq kM^2$$

and

$$E[\tilde{X}_n | \tilde{X}_{n-1} = k] = kE\tilde{\xi}_{1,1} \geq k.$$

Therefore,

$$P\left\{\tilde{X}_n \leq \left(1 - \frac{\varepsilon}{4}\right) E[\tilde{X}_n | \tilde{X}_{n-1} = k] \middle| \tilde{X}_{n-1} = k\right\} \leq \frac{16M^2}{\varepsilon^2 k}$$

and thus

$$P\left\{\tilde{X}_n \leq \left(1 + \frac{\varepsilon}{2}\tilde{X}_{n-1}\right) \middle| \mathcal{F}_{n-1}\right\} \leq \frac{16M^2}{\varepsilon^2 \tilde{X}_{n-1}}$$

Define $n_0 = \lceil 64M^2\varepsilon^{-3} \rceil$, and event $B_n = \left\{\tilde{X}_n \geq \left(1 + \frac{\varepsilon}{2}\right)^{n-n_0} 64M^2\varepsilon^{-3}\right\}$ for $n \geq n_0$. Since $E\tilde{\xi}_{1,1} > 1$, $P(B_{n_0}) > 0$. Also It follows that

$$P(B_{n_0+1}|B_{n_0}) \geq 1 - \frac{16M^2}{\varepsilon^2 \cdot 64M^2\varepsilon^{-3}} = 1 - \frac{\varepsilon}{4}.$$

Similarly, we have $P(B_{n_0+2}|B_{n_0+1} \cap B_{n_0}) \geq 1 - \frac{1}{4}\left(1 + \frac{\varepsilon}{2}\right)^{-1}\varepsilon$, etc. Therefore, for the event $C = \left\{\tilde{X}_n \geq \left(1 + \frac{\varepsilon}{2}\right)^{n-n_0} \tilde{X}_{n_0} \text{ for all } n > n_0 \text{ and } \tilde{X}_{n_0} \geq 64M^2\varepsilon^{-3}\right\}$, we have

$$P(C) \geq P\left\{\tilde{X}_{n_0} \geq 64M^2\varepsilon^{-3}\right\} \prod_{n=n_0+1}^{\infty} \left(1 - \frac{1}{4}\left(1 + \frac{\varepsilon}{2}\right)^{n_0-n}\varepsilon\right) > 0.$$

5.8 Unifrom integrability, convergence in L^1

Theorem (uniform integrability). Let $\{X_n\}_{n=1}^{\infty}$ are random variables with $E|X_n| < \infty$. Suppose X is another random variable such that X_n converges to X in probability. Then the following are equivalent:

1. $X_n \rightarrow X$ in L^1 .
2. $\{X_n\}_{n=1}^{\infty}$ is uniformly integrable, i.e.

$$\lim_{M \rightarrow \infty} \sup_n E[|X_n| \mathbf{1}_{\{|X_n| > M\}}] = 0.$$

Proof. (1) $\implies$ (2). Suppose $X_n \rightarrow X$ in L^1 , then $E|X| < \infty$ and $s = \sup_n E|X_n| < \infty$. Fix $\varepsilon > 0$, then there exists $\delta > 0$ such that $A \in \Sigma$ with $P(A) \leq \delta$ implies $E[|X| \mathbf{1}_A] < \varepsilon$. Let n_0 be such that $n \geq n_0$ implies $E|X - X_n| \leq \varepsilon$. There also exists $M_0 < \infty$ such that

$$\max_{n < n_0} E[|X_n| \mathbf{1}_{\{|X_n| \geq M_0\}}] < \varepsilon.$$

Now let $M = \max(M_0, \delta^{-1}s)$. Then for $n \geq n_0$, we have

$$E[|X_n| \mathbf{1}_{\{|X_n| \geq M\}}] \leq E|X - X_n| + E[|X| \mathbf{1}_{\{|X_n| \geq M\}}] \leq 2\varepsilon,$$

where for the second term we used Markov inequality to obtain $P\{|X_n| \geq M\} \leq M^{-1}s \leq \delta$.

(2) $\implies$ (1). Suppose X_n does not converge to X in L^1 . Then there is a subsequence $\{X_{n(k)}\}_{k=1}^{\infty}$ and $\varepsilon > 0$ such that $E|X_{n(k)} - X| \geq \varepsilon$ for all k . To simplify notation without loss of generality, consider X_n with $E|X_n - X| \geq \varepsilon$ for all n .

In the case where $E|X| = \infty$, with $X_n \xrightarrow{P} X$ claim that $\sup_n E|X_n| = \infty$. Indeed, since $X_n \rightarrow X$ in probability, there is a subsequence $X_{n(k)}$ such that $X_{n(k)} \rightarrow X$ a.s. If $\sup_n E|X_n|$ is finite, Fatou's lemma then gives

$$E|X| \leq \liminf E|X_n| < \infty,$$

a contradiction. Therefore, $\sup_n E|X_n| = \infty$ and X_n is not uniformly integrable.

Now suppose $E|X| < \infty$. Fix $\varepsilon > 0$. There is $\delta > 0$ such that $P(A) \leq \delta$ implies $E[|X_n| \mathbf{1}_A] \leq \frac{\varepsilon}{4}$. Since $X_n - X \xrightarrow{P} 0$, we have $P\{|X_n - X| \geq 1\} \rightarrow 0$ and thus

$$E[|X_n - X| \mathbf{1}_{\{|X_n - X| < 1\}}] \leq P\{|X_n - X| \geq 1\} \rightarrow 0$$

as $n \rightarrow \infty$. Now for n large enough,

$$\begin{aligned} E[|X_n| \mathbf{1}_{\{|X - X_n| \geq 1\}}] + \frac{\varepsilon}{4} &\geq E[|X_n| \mathbf{1}_{\{|X - X_n| \geq 1\}}] + E[|X| \mathbf{1}_{\{|X - X_n| \geq 1\}}] \\ &\geq E[|X - X_n| \mathbf{1}_{\{|X_n - X| \geq 1\}}] \\ &\geq E|X - X_n| - \frac{\varepsilon}{4} \\ &\geq \frac{3\varepsilon}{4}. \end{aligned}$$

Let $P\{|X_n - X| \geq 1\} = \delta_n$, then $\delta_n \rightarrow 0$. We then have

$$\liminf_{n \rightarrow \infty} E[|X_n| \mathbf{1}_{\{|X_n| > \delta_n^{-1/2}\}}] \geq \liminf_{n \rightarrow \infty} E[|X_n| \mathbf{1}_{\{|X_n| > \delta_n^{-1/2}\}} \mathbf{1}_{\{|X_n - X| \geq 1\}}].$$

However,

$$\begin{aligned} &E[|X_n| \mathbf{1}_{\{|X_n| > \delta_n^{-1/2}\}} \mathbf{1}_{\{|X_n - X| \geq 1\}}] \\ &= E[|X_n| \mathbf{1}_{\{|X - X_n| \geq 1\}}] - E[|X_n| \mathbf{1}_{\{|X_n| \leq \delta_n^{-1/2}\}} \mathbf{1}_{\{|X_n - X| \geq 1\}}] \\ &\geq \frac{\varepsilon}{2} - \delta_n^{1/2}. \end{aligned}$$

Therefore, $\liminf_n E[|X_n| \mathbf{1}_{\{|X_n| > \delta_n^{-1/2}\}}] \geq \frac{\varepsilon}{2}$ and thus $\{X_n\}_{n=1}^\infty$ is not uniformly integrable. $\square$

Theorem. Let X be a random variable with $E|X| < \infty$ and $\{\mathcal{F}_n\}_{n=1}^\infty$ be a filtration. Then, the sequence $\{E[X|\mathcal{F}_n]\}_{n=1}^\infty$ is uniformly integrable.

Proof. Fix $\varepsilon > 0$. Let $\delta > 0$ be such that for any event A with $P(A) \leq \delta$, we have $E[|X| \mathbf{1}_A] < \varepsilon$. Also, let $M = \delta^{-1} E|X|$. Claim that for any n , we have

$$E[|E[X|\mathcal{F}_n]| \mathbf{1}_{\{|E[X|\mathcal{F}_n]| \geq M\}}] \leq \varepsilon.$$

Write event $A_M = \{|E[X|\mathcal{F}_n]| \geq M\}$. First of all, from Jensen's inequality, we have

$$|E[X|\mathcal{F}_n]| \mathbf{1}_{A_M} \leq E[|X| |\mathcal{F}_n] \mathbf{1}_{A_M}$$

Note that $\mathbf{1}_{A_M}$ is $\mathcal{F}_n$ -measurable, so

$$E[E[|X| |\mathcal{F}_n] \mathbf{1}_{A_M}] = E[E[|X| \mathbf{1}_{A_M} |\mathcal{F}_n]] = E[|X| \mathbf{1}_{A_M}].$$

With Markov's inequality, we know $P(A_M) \leq \delta$. By the way δ is chosen, the proof is complete. $\square$

Theorem. Let X be a random variable with $E|X| < \infty$ and $\{\mathcal{F}_n\}_{n=1}^\infty$ be a filtration. Also write $\mathcal{F}_\infty = \sigma(\mathcal{F}_n : n \geq 1)$. Then $E[X|\mathcal{F}_n]$ converges to X a.e. and in L^1 .

Proof. Towards applying submartingale convergence theorem, we have

$$E[E[X|\mathcal{F}_n]^+] \leq E[|E[X|\mathcal{F}_n]|] \leq E|X|$$

from Jensen's inequality. By submartingale convergence theorem, there exists random variable Z with $E|X| < \infty$ such that $E[X|\mathcal{F}_n] \rightarrow Z$ a.e., which implies $E[X|\mathcal{F}_n] \rightarrow Z$ in probability. Together with previous theorems, we also have $E[X|\mathcal{F}_n] \rightarrow Z$ in L^1 .

Now the goal is to show that $Z = E[X|\mathcal{F}_\infty]$ a.e. Let event $A \in \mathcal{F}_m$, then for any $n \geq m$, $\mathbf{1}_A E[X|\mathcal{F}_n] = E[X \mathbf{1}_A |\mathcal{F}_n]$. This implies $E[\mathbf{1}_A E[X|\mathcal{F}_n]] = E[\mathbf{1}_A X]$. Now, $E[X|\mathcal{F}_n] \rightarrow Z$ in L^1 implies $E[\mathbf{1}_A E[X|\mathcal{F}_n]] \rightarrow E[\mathbf{1}_A Z]$. Therefore, $E[\mathbf{1}_A X] = E[\mathbf{1}_A Z]$ for any $A \in \bigcup_{m=1}^\infty \mathcal{F}_m$. Note that $\bigcup_{m=1}^\infty \mathcal{F}_m$ is a π -system that generates $\mathcal{F}_\infty$. We can also check that all sets $A \in \mathcal{F}_\infty$ that satisfy $E[\mathbf{1}_A X] = E[\mathbf{1}_A Z]$ is a λ -system. Applying π - λ theorem and recall the definition of conditional expectation completes the proof. $\square$

Corollary (Levy 0-1 law). Let $\{\mathcal{F}_n\}_{n=1}^\infty$ be a filtration and let $\mathcal{F}_\infty = \sigma(\mathcal{F}_n : n \geq 1)$. Then, $E[\mathbf{1}_A |\mathcal{F}_n] \rightarrow \mathbf{1}_A$ a.e. for any $A \in \mathcal{F}_\infty$.

Remark. This implies Kolmogorov 0-1 law. Indeed, suppose $\{X_n\}_{n=1}^\infty$ are mutually independent and let $\mathcal{T} = \bigcap_{m=1}^\infty \sigma(X_n : n \geq m)$ be the tail σ -field. Also let $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$. Note that $\mathcal{T} \subset \mathcal{F}_\infty$. Now take $A \in \mathcal{T}$, then A is independent of any $\mathcal{F}_n$. Therefore $E[\mathbf{1}_A |\mathcal{F}_n] = P(A)$. By Levy 0-1 law, we can conclude that $P(A) \in \{0, 1\}$.

Example. Let $\{Y_n\}_{n=1}^\infty$ be mean zero, mutually independent random variables. Immediately $\{Y_n\}_{n=1}^\infty$ is a martingale. Suppose also $\sum_{n=1}^\infty EY_n^2 < \infty$. Define $S_n = \sum_{k=1}^n Y_k$, then

$$E[S_n^+] \leq E|S_n| \leq (E|S_n|^2)^{1/2}$$

by Jensen's inequality. Now submartingale convergence theorem gives a random variable X with $E|X| < \infty$ such that $\sum_{k=1}^\infty Y_k = X$ a.e. As an exercise, we can verify that $E[X|Y_1, \dots, Y_n] = \sum_{k=1}^n Y_k$. Now, by previous theorems, we have $\sum_{k=1}^\infty Y_k \rightarrow X$ in L^1 .

5.9 Backward martingales

Definition (backward martingale). Let $\{\mathcal{F}_n\}_{n=-\infty}^0$ be a filtration such that $\mathcal{F}_{-k} \subset \mathcal{F}_{-k+1}$ for all $k \geq 1$. Let $\{X_n\}_{n=-\infty}^0$ be adapted to the filtration. Suppose $E|X_n| < \infty$ for all $n \leq 0$. If $E[X_{n+1}|\mathcal{F}_n] = X_n$ for all $n \leq -1$, then say $\{X_n\}_{n=-\infty}^0$ is a *backward martingale*.

Theorem. Let $\{X_n\}_{n=-\infty}^0$ be a backward martingale under the filtration $\{\mathcal{F}_n\}_{n=-\infty}^0$. Write $\mathcal{F}_{-\infty} = \bigcap_{m=-\infty}^0 \mathcal{F}_m$, then $X_m \rightarrow E[X_0|\mathcal{F}_{-\infty}]$ a.e. as $m \rightarrow -\infty$.

Proof. For each $m \leq -1$, define sequence $\{X_{n \wedge 0}\}_{n=m}^\infty$. This is a martingale. Fix $a < b$ and let $\{U_{m,a,b}\}_{m=-1}^{-\infty}$ be the number of upcrossings. By upcrossing lemma, we have

$$(b-a)E[U_{m,a,b}] \leq E[(X_0 - a)^+].$$

We can check that $\{U_{m,a,b}\}_{m=-1}^{-\infty}$ is monotone everywhere on Ω . This allows us to define

$$U_{-\infty,a,b} = \lim_{m \rightarrow -\infty} U_{m,a,b}.$$

We have $E[U_{-\infty,a,b}] < \infty$. The rest of the argument is similar to martingale convergence theorem, and we can show that $\liminf_{m \rightarrow -\infty} X_m = \limsup_{m \rightarrow -\infty} X_m$. That is, there exists an a.e. limit $\lim_{m \rightarrow -\infty} X_m = X_{-\infty}$ with $E|X_{-\infty}| < \infty$.

It remains to show that $X_{-\infty} = E[X_0|\mathcal{F}_{-\infty}]$. With our construction, we have $X_n = E[X_0|\mathcal{F}_n]$. This implies that X_n 's are uniformly integrable, and thus $X_n \rightarrow X_{-\infty}$ in L^1 . With L^1 convergence, for any event A we have $E[\mathbf{1}_A X_n] \rightarrow E[\mathbf{1}_A X_{-\infty}]$. Note that

$$E[\mathbf{1}_A X_n] = E[\mathbf{1}_A E[X_0|\mathcal{F}_n]] = E[\mathbf{1}_A X_0]$$

for any $A \in \mathcal{F}_{-\infty}$. It follows that $E[\mathbf{1}_A X_0] = E[\mathbf{1}_A X_{-\infty}]$. Since $X_{-\infty}$ is $\mathcal{F}_{-\infty}$ -measurable, we can conclude that $X_{-\infty} = E[X_0|\mathcal{F}_{-\infty}]$. □

With this, we can revisit law of large numbers.

Theorem (strong law of large numbers). Let $X_1, X_2, \dots$ be i.i.d. random variables with $EX_1 = 0$. Define $S_n = \sum_{i=1}^n X_i$. We then have $S_n/n \rightarrow 0$ a.e.

Proof. Define random variable $Y_{-n} = S_n$ and filtration

$$\mathcal{F}_{-n} = \sigma(Y_{-n}, Y_{-n-1}, Y_{-n-2}, \dots) = \sigma\left(\sum_{i=1}^n X_i, X_{n+1}, X_{n+2}, \dots\right).$$

That is, we don't have precise information of the first n elements. Claim that $\{Y_n\}_{n=-\infty}^{-1}$ is a backward martingale, i.e. $E[Y_{-n+1}|\mathcal{F}_{-n}] = Y_{-n}$ for all $n \leq -1$. This means we need to show that

$$E\left[\frac{1}{n-1} \sum_{i=1}^{n-1} X_i \middle| \sum_{i=1}^n X_i, X_{n+1}, \dots\right] = \frac{1}{n} \sum_{i=1}^n X_i.$$

Let $A \in \sigma(\sum_{i=1}^n X_i, X_{n+1}, X_{n+2}, \dots)$. This means $A = \{(\sum_{i=1}^n X_i, X_{n+1}, X_{n+2}, \dots) \in B\}$ for some Borel set B . Since X_i 's are i.i.d. and A only depends on $X_1, \dots, X_n$ through $\sum_{i=1}^n X_i$, by symmetry we must have

$$P(A \cap \{X_i \in C\}) = P(A \cap \{X_j \in C\})$$

for any $1 \leq i, j \leq n$ and any $C \subset \mathcal{B}_{\mathbb{R}}$. Therefore, $E[X_i \mathbf{1}_A]$ is the same for all $1 \leq i \leq n-1$ and thus $E[\sum_{i=1}^{n-1} X_i \mathbf{1}_A] = (n-1)E[X_1 \mathbf{1}_A]$. Similarly, $E[\sum_{i=1}^n X_i \mathbf{1}_A] = nE[X_1 \mathbf{1}_A]$. This proves the equation and confirms that Y_n is a backward martingale.

With the previous theorem, we have $Y_{-n} \rightarrow E[Y_{-1} | \mathcal{F}_{-\infty}]$ a.e. However, the weak law of large numbers gives $Y_{-n} \rightarrow 0$ in probability, so we must have $Y_{-n} \rightarrow 0$ a.e.

□